Electronic Devices and Circuit Theory

Today's Topics:

- Diodes and its applications.

Announcement!

- **Quiz 1 date:**

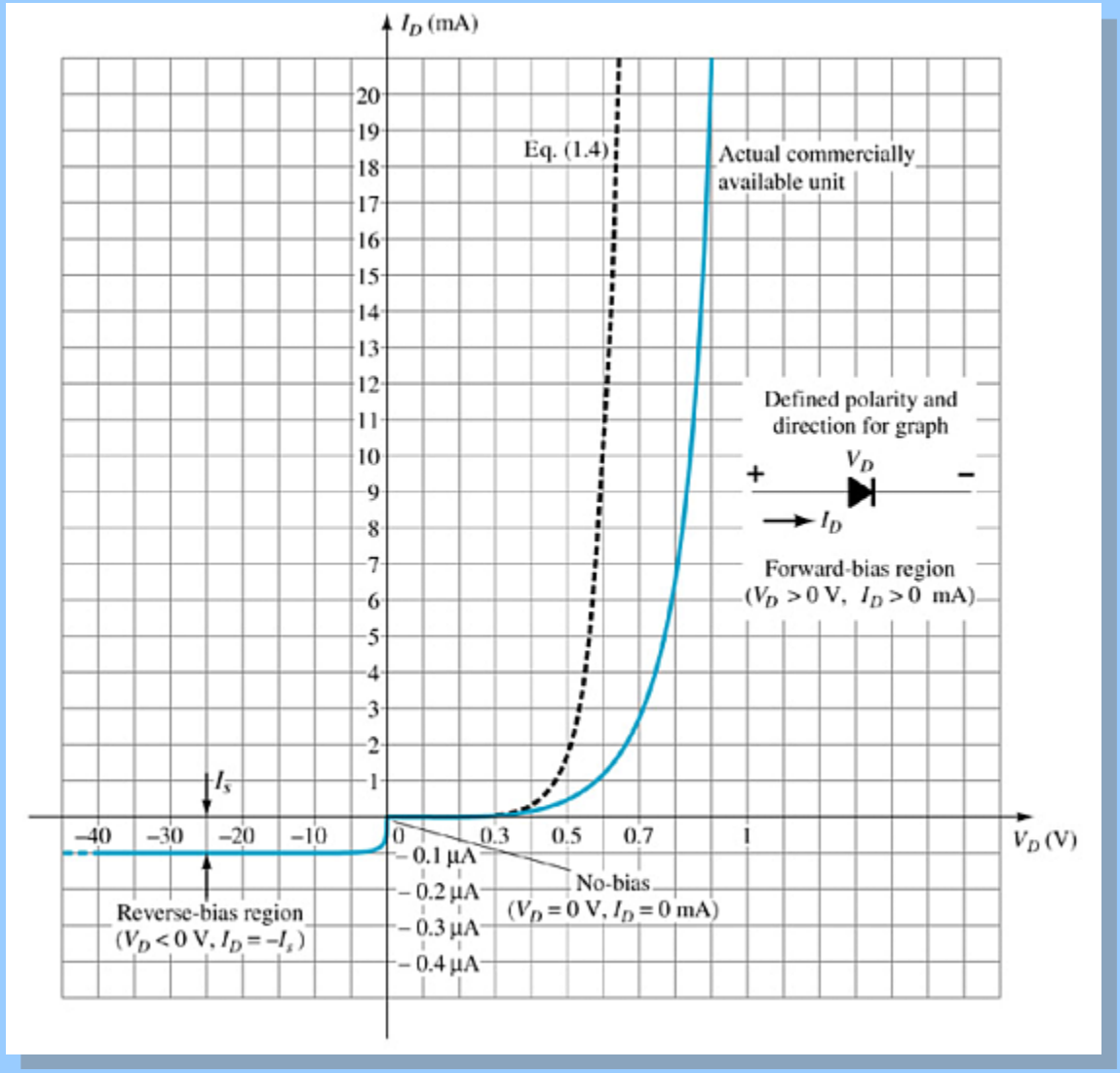
26th March – at 5pm

Classrooms: Z11 and 103

Actual Diode Characteristics

Note the regions for no bias, reverse bias, and forward bias conditions.

Carefully note the scale for each of these conditions.



Actual Diode Characteristics

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad \text{(A)} \tag{1.2}$$

where I_s is the reverse saturation current
V_D is the applied forward-bias voltage across the diode
n is an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ($n = 1$ will be assumed throughout this text unless otherwise noted).

The voltage V_T in Eq. (1.1) is called the *thermal voltage* and is determined by

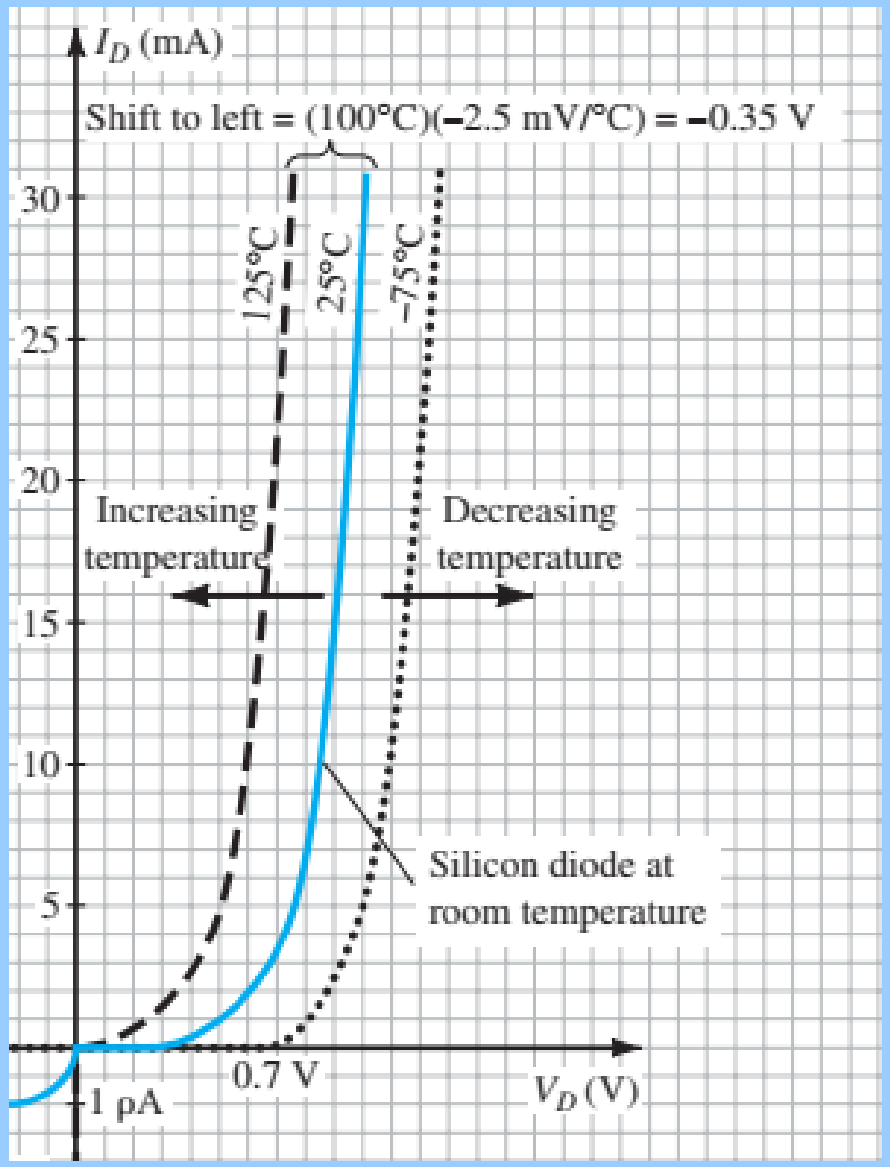
$$V_T = \frac{kT_K}{q} \quad \text{(V)} \tag{1.3}$$

where k is Boltzmann's constant $= 1.38 \times 10^{-23}$ J/K
T_K is the absolute temperature in kelvins = 273 + the temperature in °C
q is the magnitude of electronic charge $= 1.6 \times 10^{-19}$ C

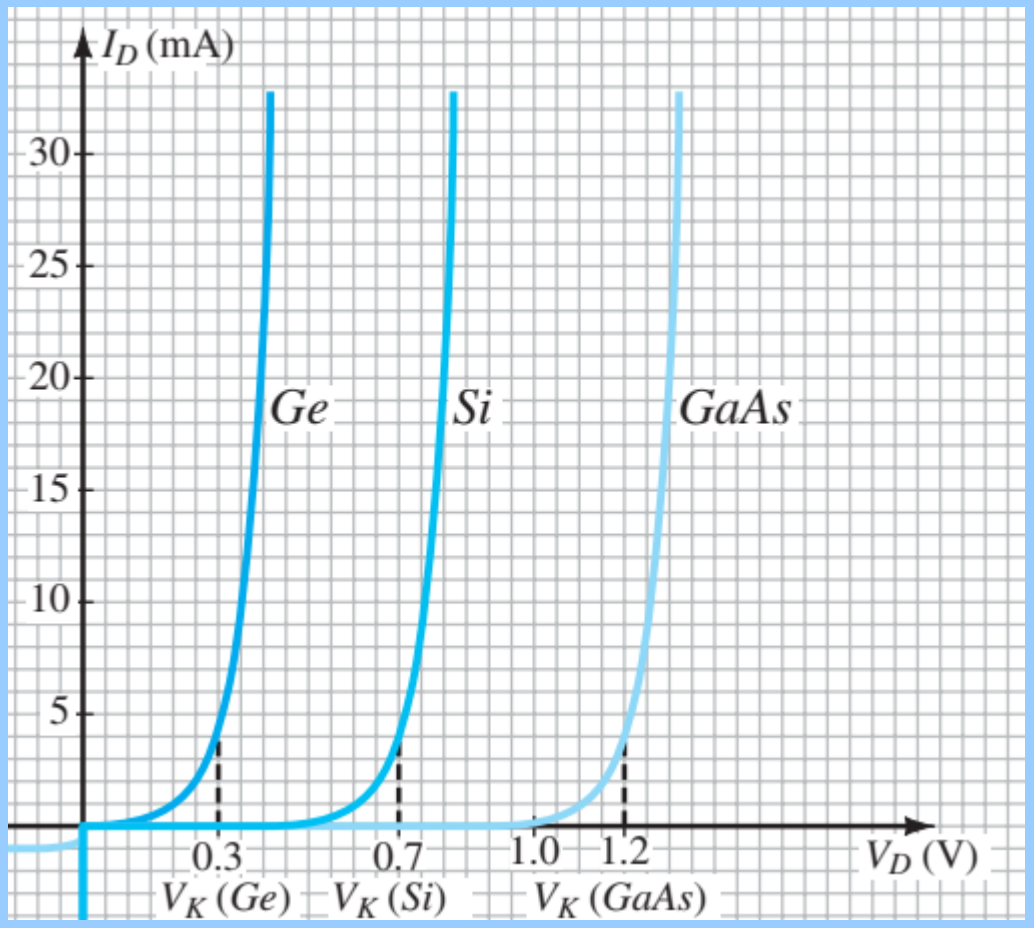
Actual Diode Characteristics

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad \text{(A)}$$

$$V_T = \frac{kT_K}{q} \quad \text{(V)}$$



Actual Diode Characteristics



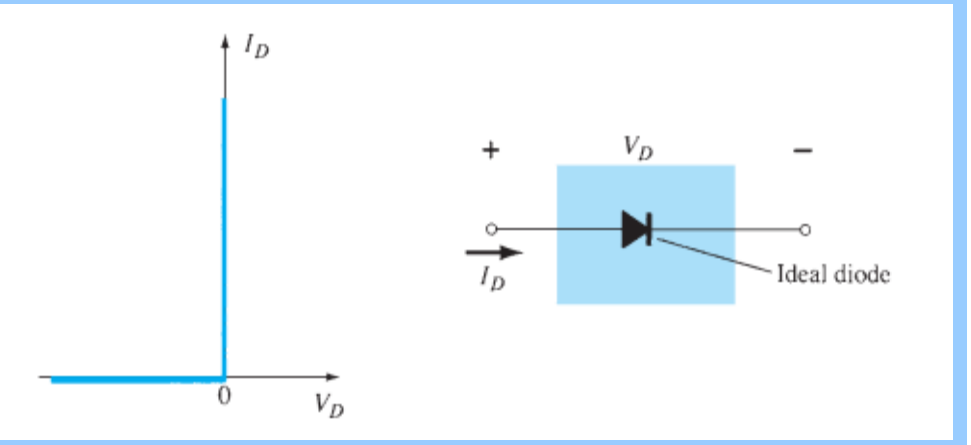
Forward Bias Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the *p-n* junction. This energy comes from the external voltage applied across the diode.

The forward bias voltage required for a:

- **gallium arsenide diode ≅ 1.2 V**
- **silicon diode ≅ 0.7 V**
- **germanium diode ≅ 0.3 V**

Diode – Ideal vs. Actual



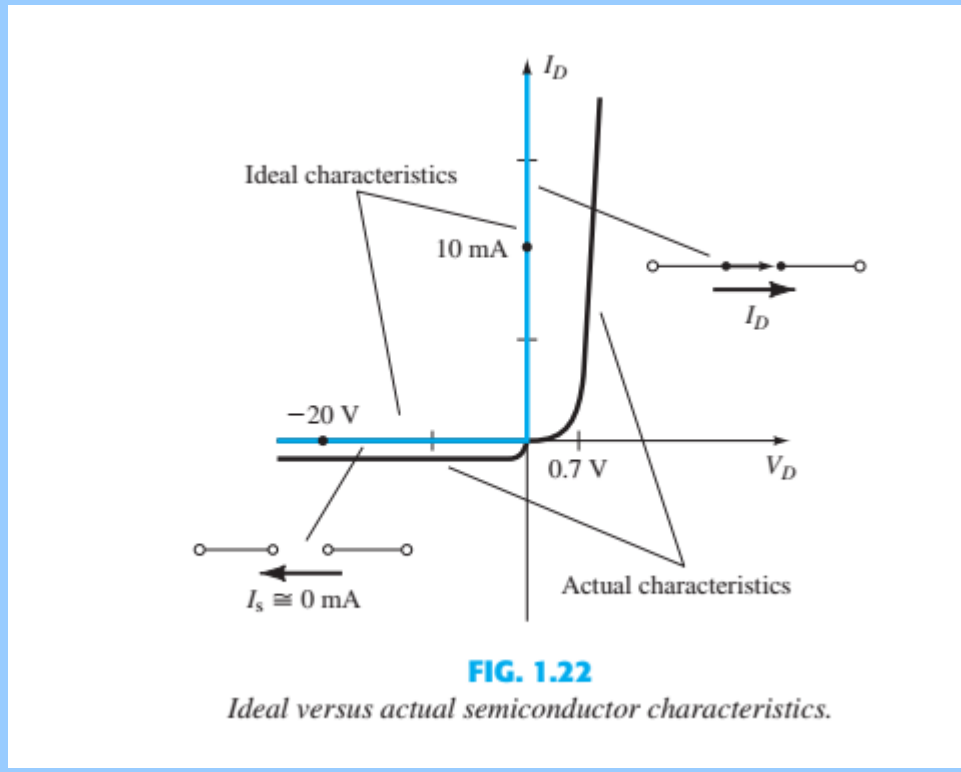
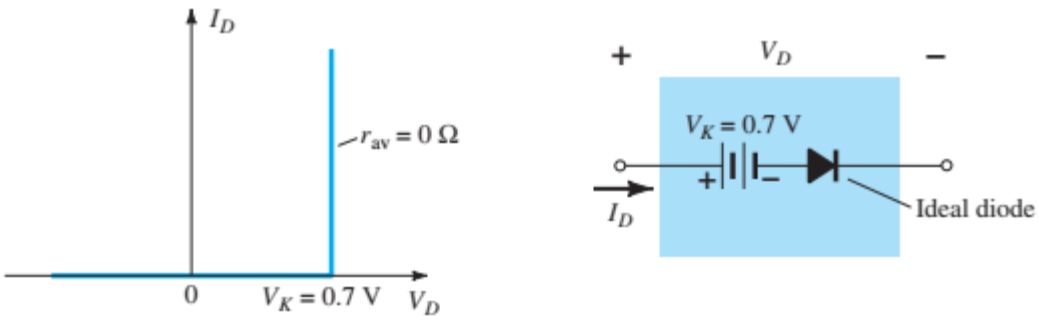


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode (Constant Voltage Model)



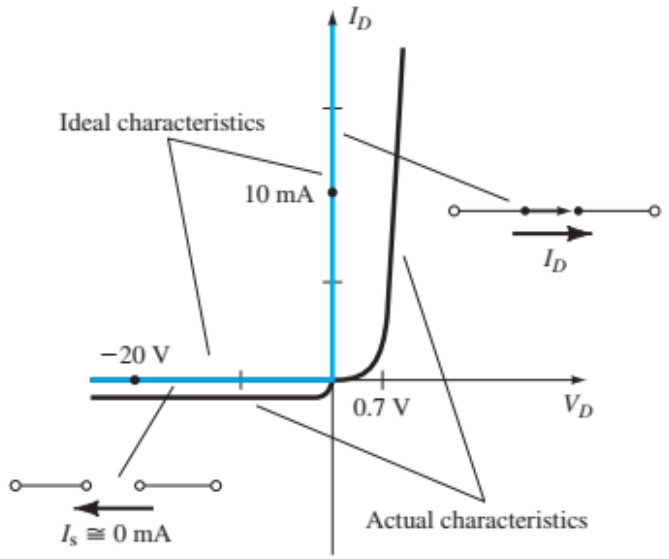


FIG. 1.22
Ideal versus actual semiconductor characteristics.

Diode (Approximation)

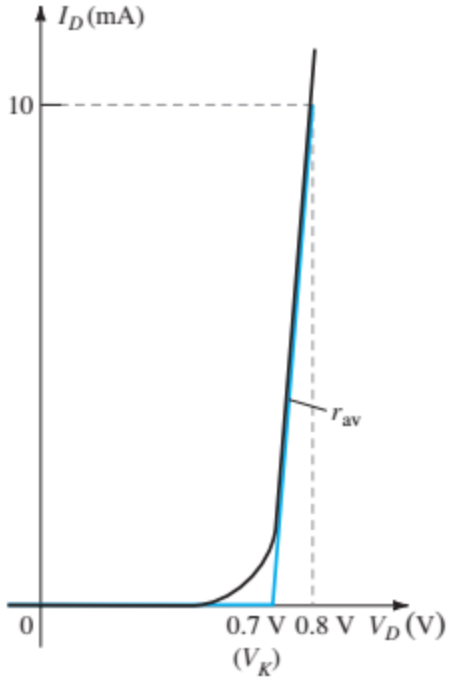
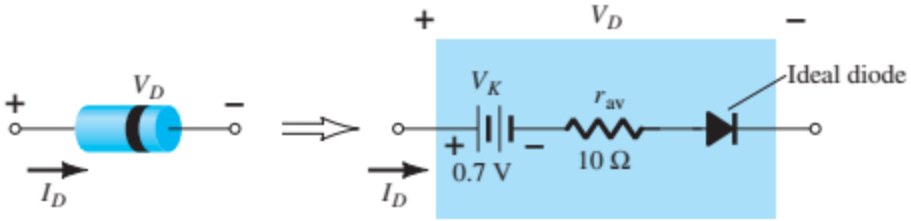


FIG. 1.29

Defining the piecewise-linear equivalent circuit using straight-line segments to approximate the characteristic curve.



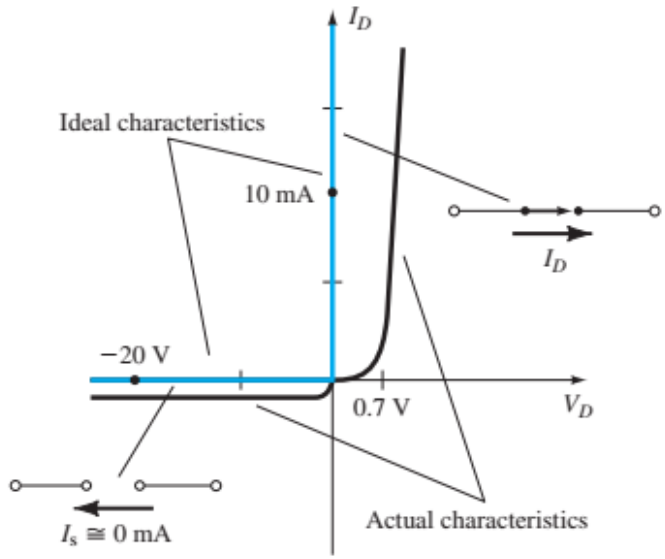


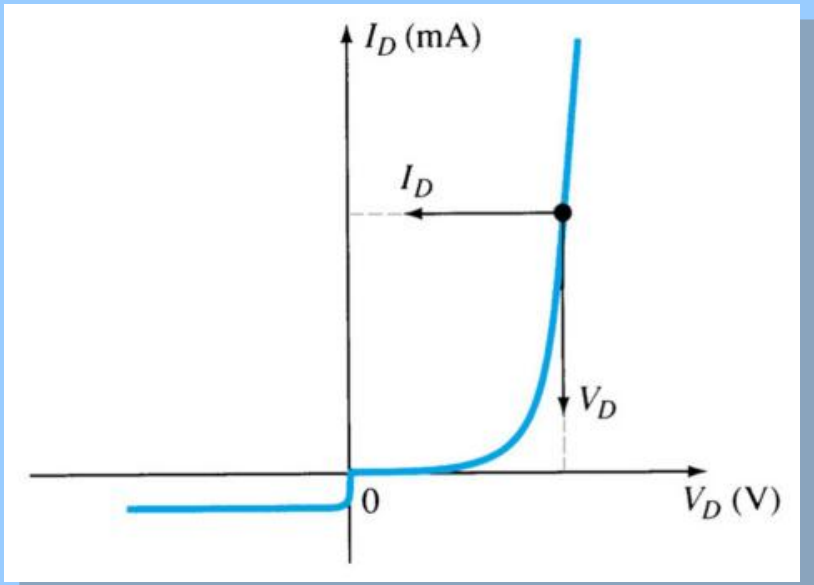
FIG. 1.22

Ideal versus actual semiconductor characteristics.

DC (Static) Resistance

For a specific applied DC voltage V_D, the diode has a specific current I_D, and a specific resistance R_D.

$$R_D = \frac{V_D}{I_D}$$



Diode – AC resistance

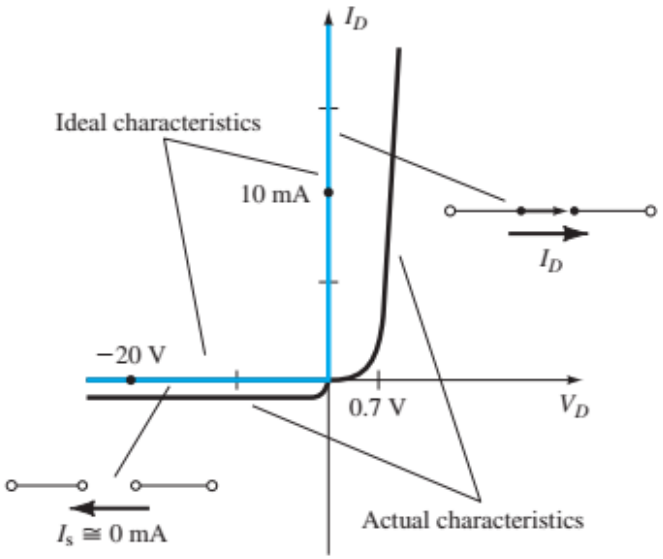
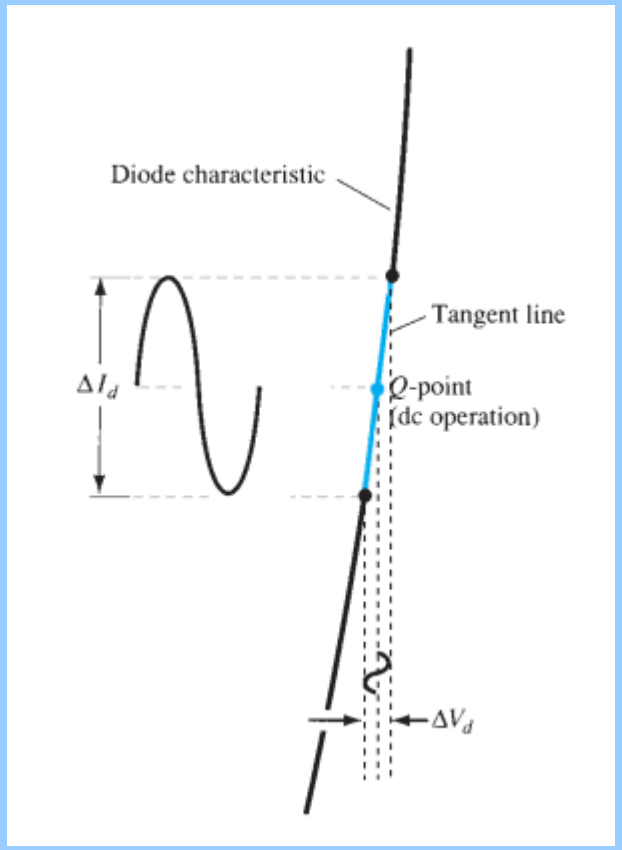
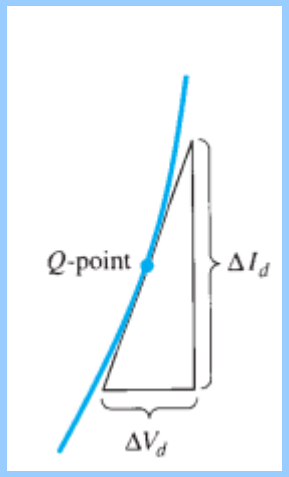


FIG. 1.22
Ideal versus actual semiconductor characteristics.



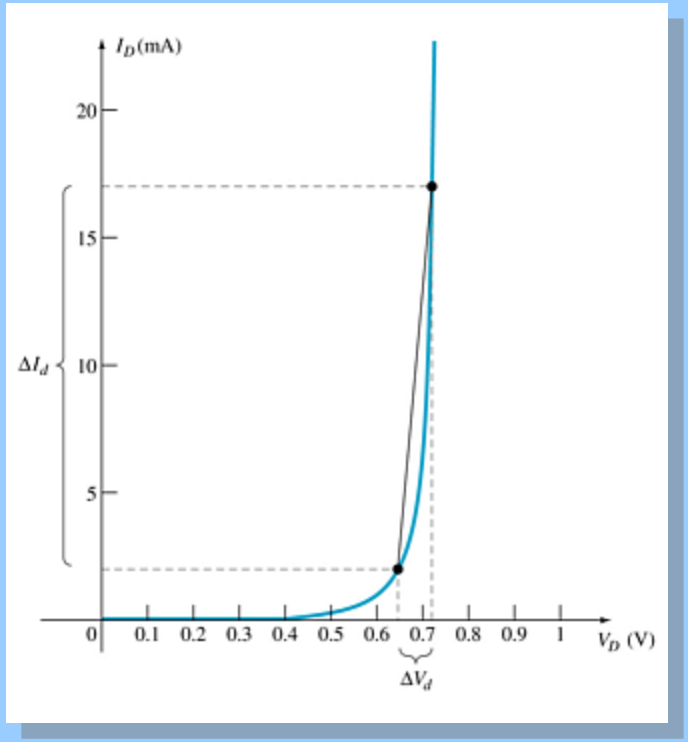
$$r_d = \frac{\Delta V_d}{\Delta I_d}$$



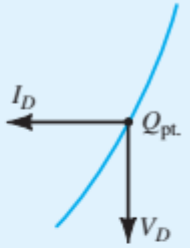
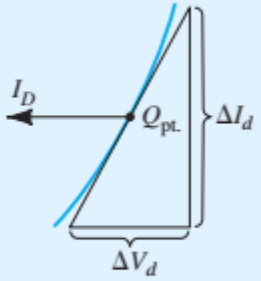
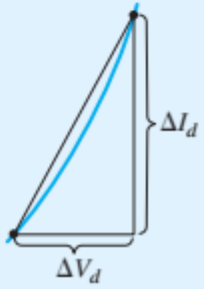
Average AC Resistance

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \Big| \text{ pt. to pt.}$$

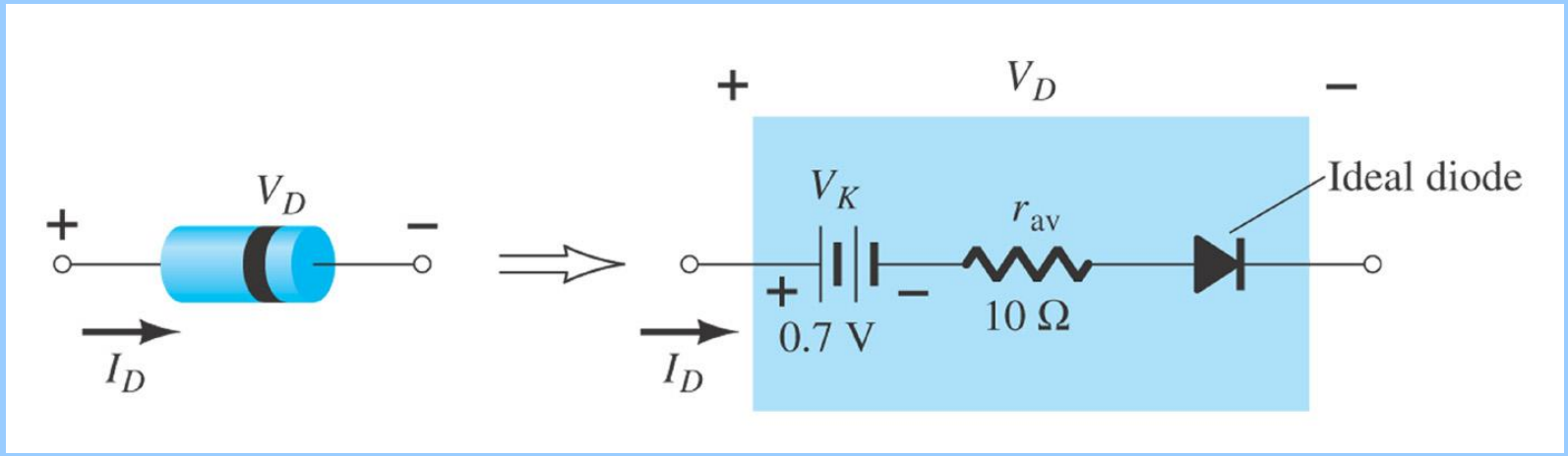
AC resistance can be calculated using the current and voltage values for two points on the diode characteristic curve.

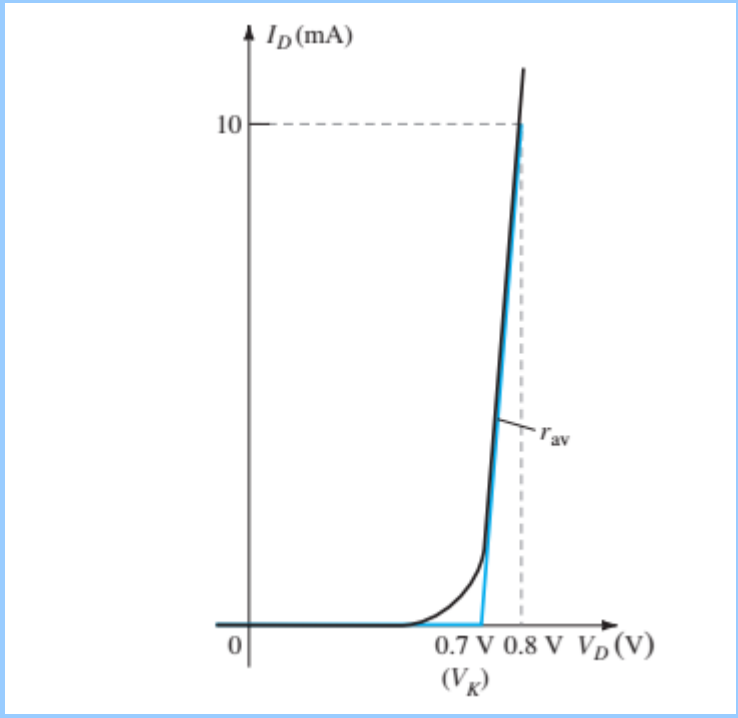


Diode – Resistance Levels

Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26\text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right\|_{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

Diode Equivalent Circuit

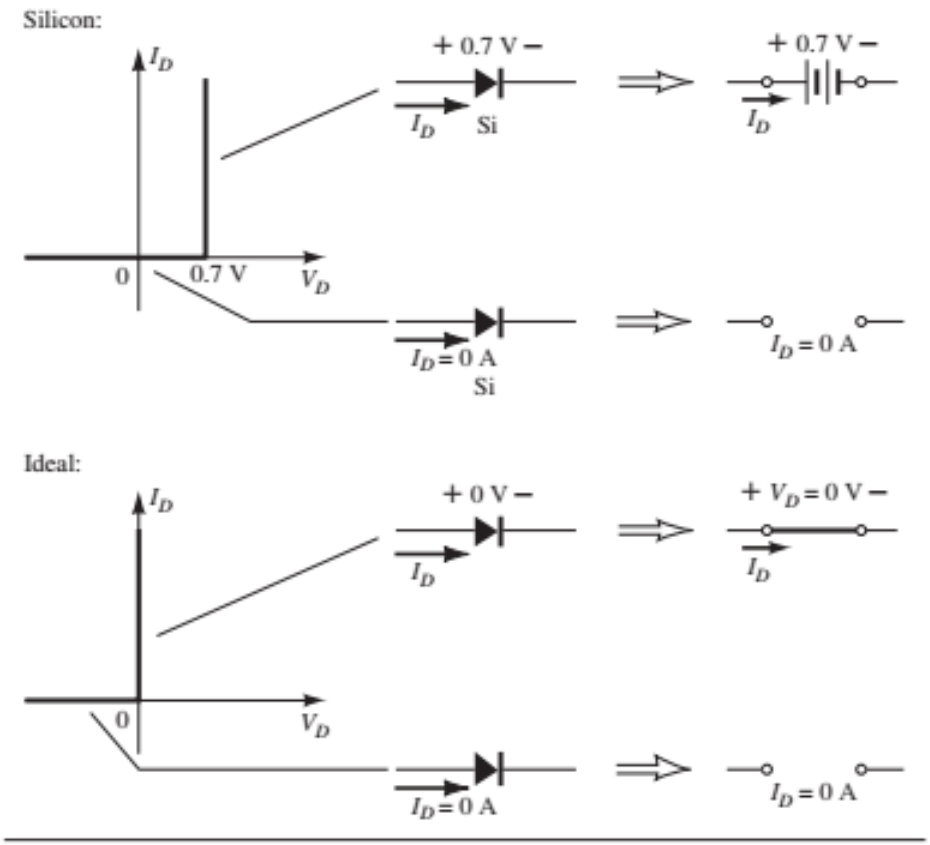




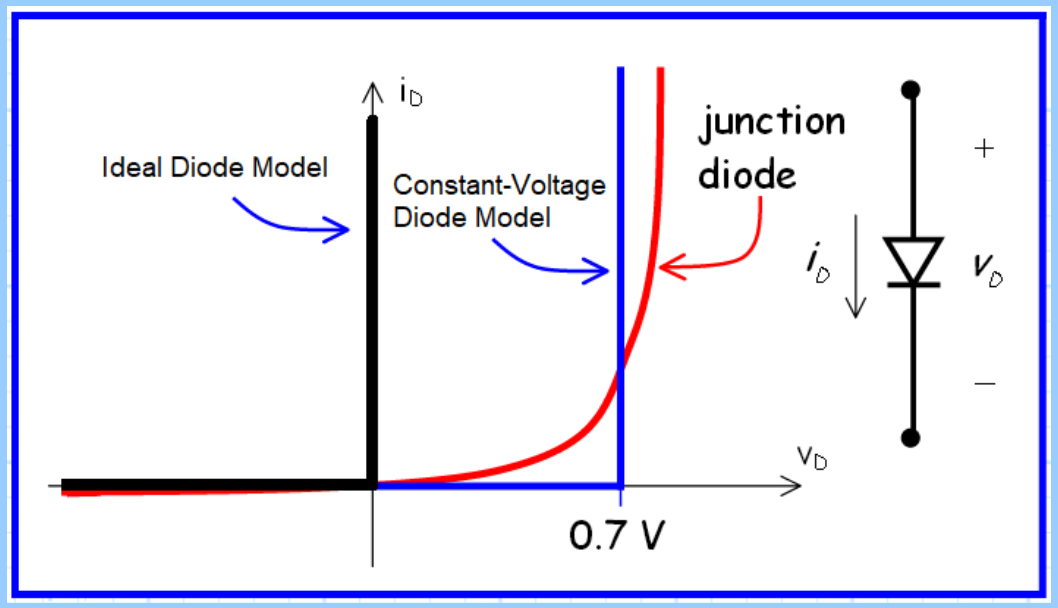
Constant Voltage Si Diode Model

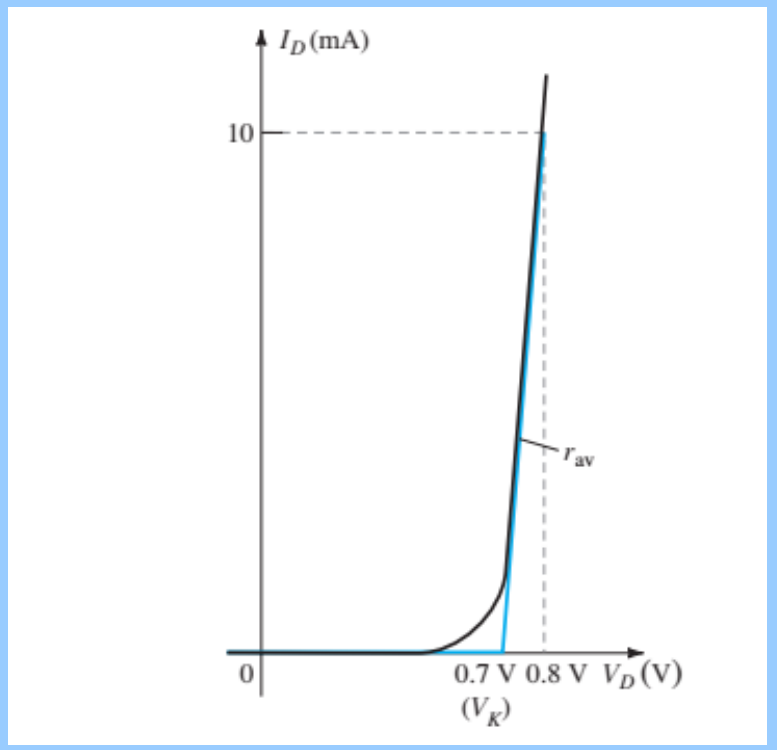
TABLE 2.1

Approximate and Ideal Semiconductor Diode Models.



Diode Models





Series Diode Configurations

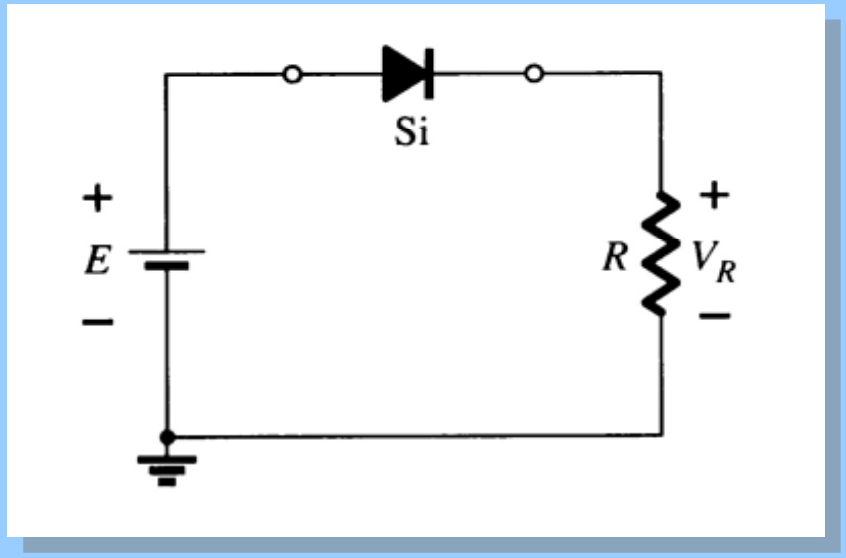
Forward Bias

Constants

- **Silicon Diode: $V_D = 0.7$ V**
- **Germanium Diode: $V_D = 0.3$ V**

Analysis (for silicon)

- $V_D = 0.7$ V (or $V_D = E$ if $E < 0.7$ V)
- $V_R = E - V_D$
- $I_D = I_R = I_T = V_R / R$



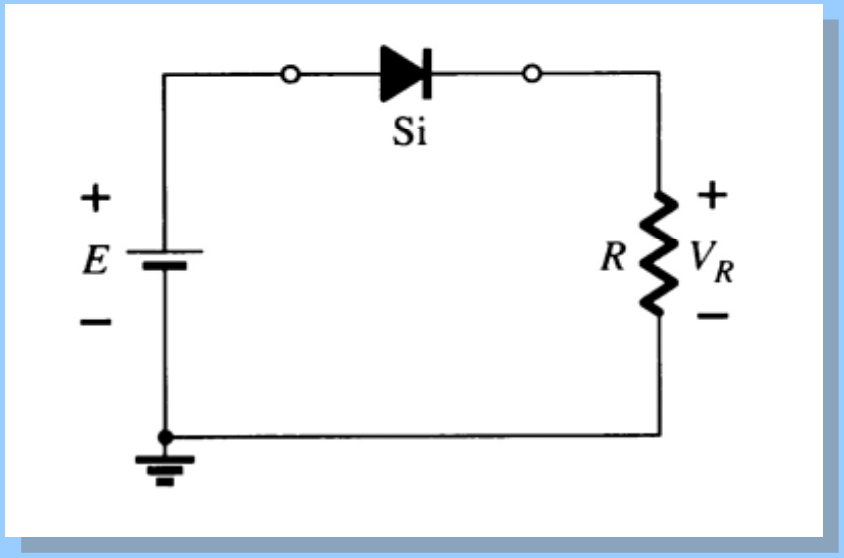
Series Diode Configurations

Reverse Bias

Diodes ideally behave as open circuits

Analysis

- $V_D = E$
- $V_R = 0\ \text{V}$
- $I_D = 0\ \text{A}$



Series Diode Configurations

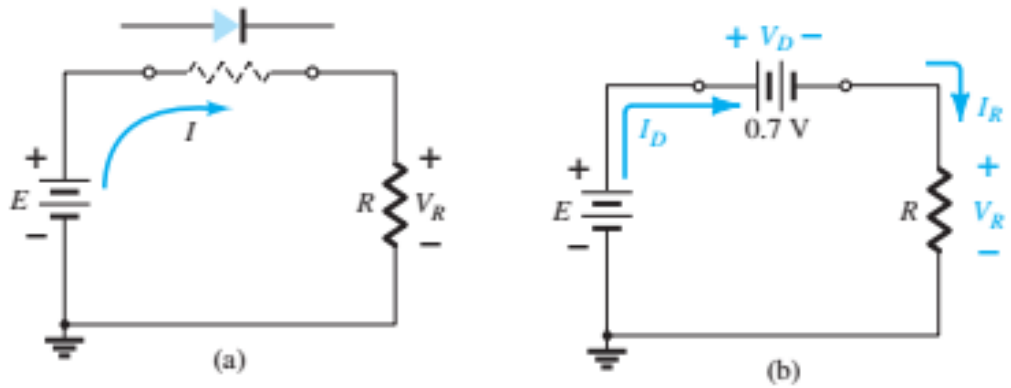


FIG. 2.9

(a) Determining the state of the diode of Fig. 2.8; (b) substituting the equivalent model for the "on" diode of Fig. 2.9a.

Series Diode Configurations

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D, V_R, and I_D.

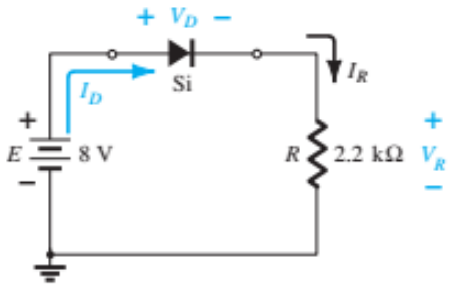


FIG. 2.13
Circuit for Example 2.4.

Series Diode Configurations

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D, V_R, and I_D.

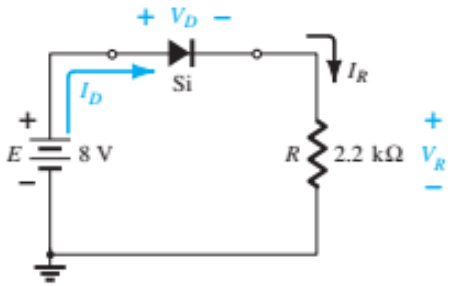


FIG. 2.13
Circuit for Example 2.4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the "on" state,

$$V_D = \mathbf{0.7\ V}$$

$$V_R = E - V_D = 8\ \text{V} - 0.7\ \text{V} = \mathbf{7.3\ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\ \text{V}}{2.2\ \text{k}\Omega} \cong \mathbf{3.32\ mA}$$

Series Diode Configurations

EXAMPLE 2.5 Repeat Example 2.4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 2.14, where $I_D = \mathbf{0\ A}$ due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = \mathbf{0\ V}$. Applying Kirchhoff's voltage law around the closed loop yields

$$E - V_D - V_R = 0$$

and

$$V_D = E - V_R = E - 0 = E = 8\ \text{V}$$

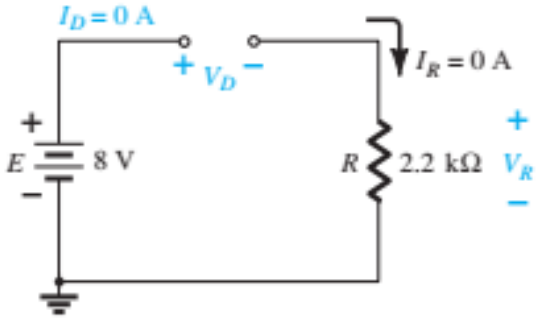


FIG. 2.14
Determining the unknown quantities for Example 2.5.

Parallel Configurations

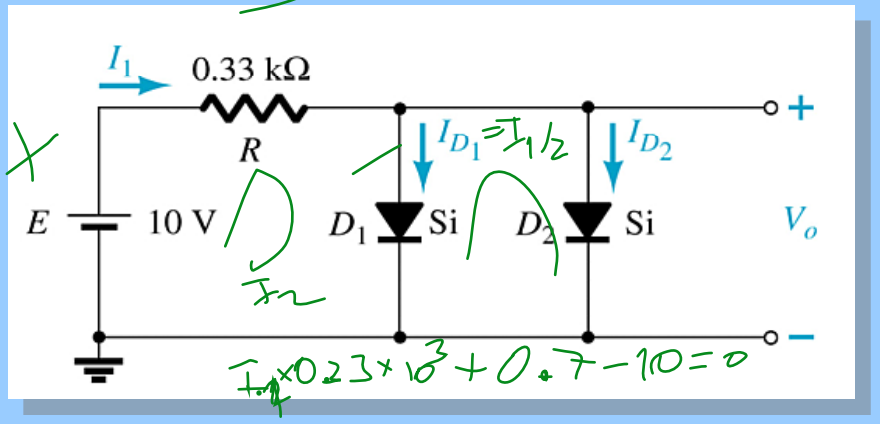
$$V_D = 0.7\,V$$

$$V_{D1} = V_{D2} = V_O = 0.7\,V$$

$$V_R = 9.3\,V$$

$$I_R = \frac{E - V_D}{R} = \frac{10\,V - .7\,V}{.33k\Omega} = 28\,mA$$

$$I_{D1} = I_{D2} = \frac{28\,mA}{2} = 14\,mA$$



Parallel Configurations

EXAMPLE 2.12 Determine the voltage V_o for the network of Fig. 2.35.

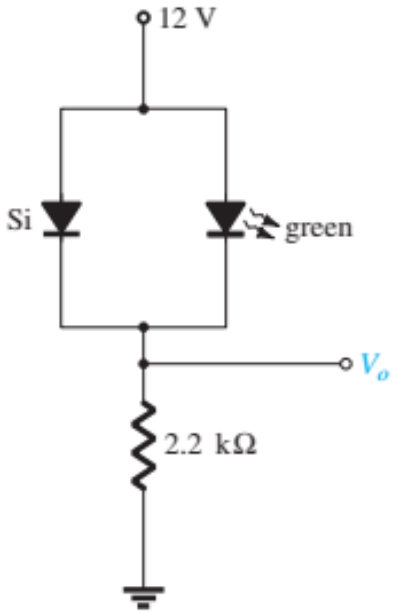


FIG. 2.35
Network for Example 2.12.

Parallel Configurations

EXAMPLE 2.12 Determine the voltage V_o for the network of Fig. 2.35.

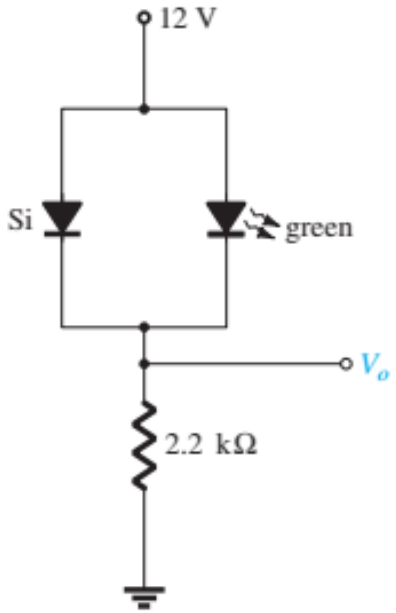


FIG. 2.35
Network for Example 2.12.

$$V_o = 12\text{ V} - 0.7\text{ V} = \mathbf{11.3\text{ V}}$$

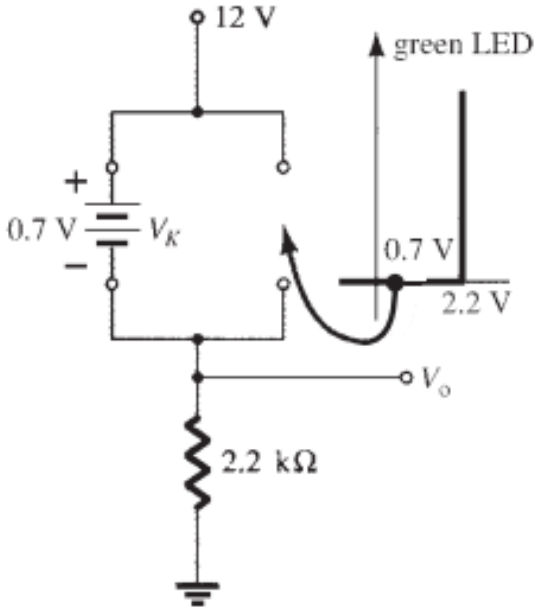


FIG. 2.36
Determining V_o for the network of Fig. 2.35.

Parallel Configurations

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

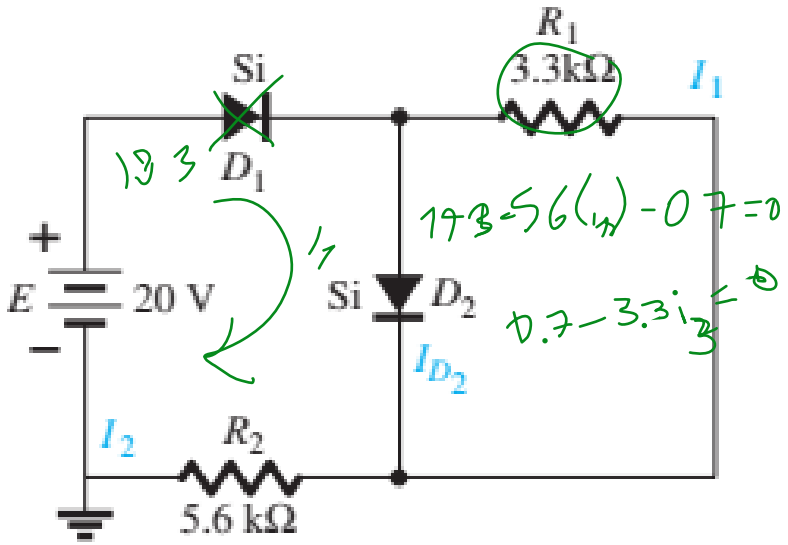


FIG. 2.37
Network for Example 2.13.

Parallel Configurations

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

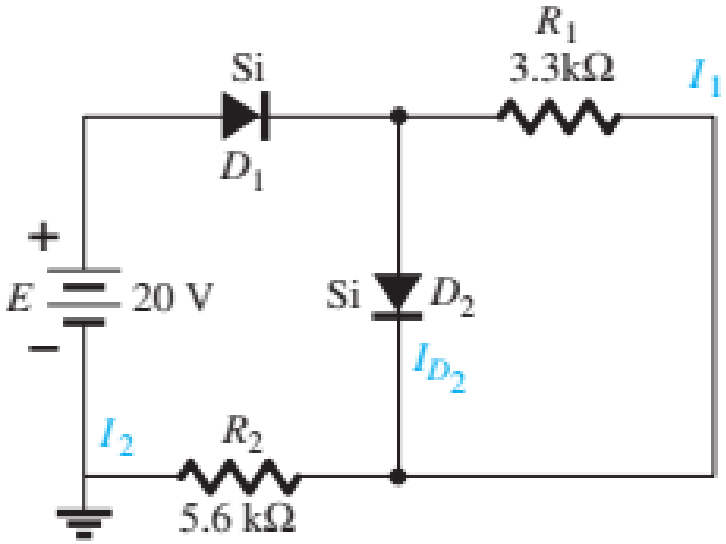


FIG. 2.37
Network for Example 2.13.

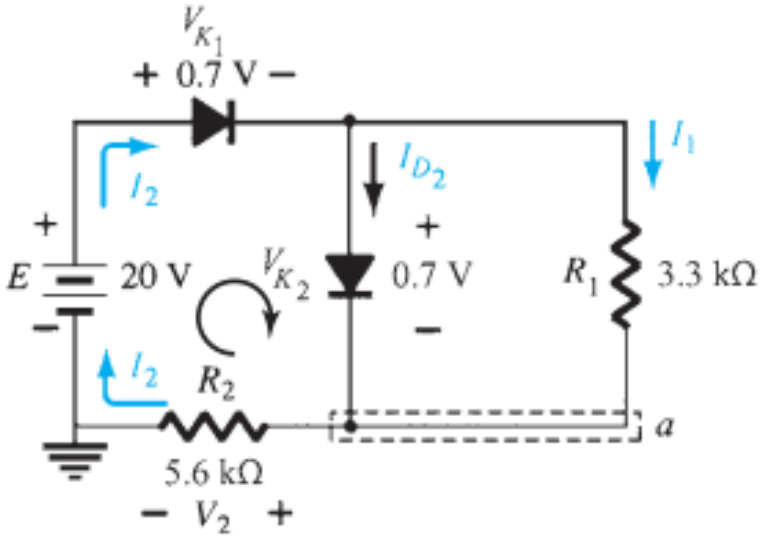


FIG. 2.38
Determining the unknown quantities for Example 2.13.

Parallel Configurations

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

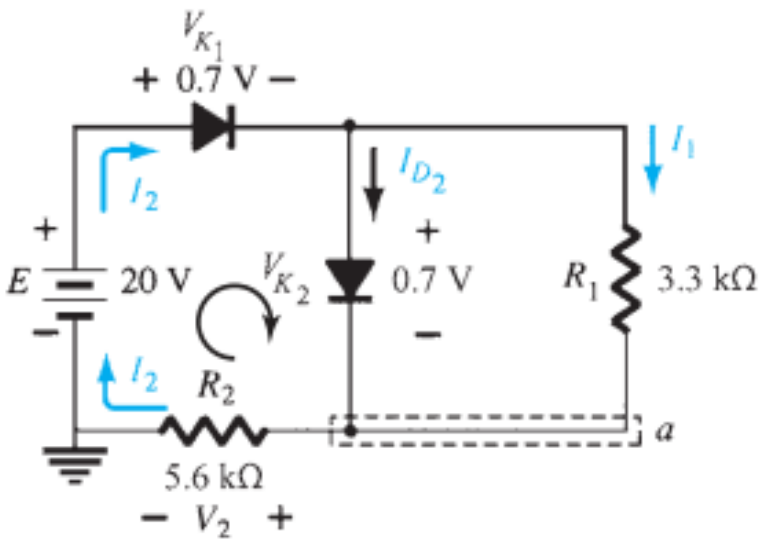


FIG. 2.38
Determining the unknown quantities for Example 2.13.

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7\text{ V}}{3.3\text{ k}\Omega} = \mathbf{0.212\text{ mA}}$$

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

$$V_2 = E - V_{K_1} - V_{K_2} = 20\text{ V} - 0.7\text{ V} - 0.7\text{ V} = \mathbf{18.6\text{ V}}$$

$$I_2 = \frac{V_2}{R_2} = \frac{18.6\text{ V}}{5.6\text{ k}\Omega} = \mathbf{3.32\text{ mA}}$$

$$I_{D_2} + I_1 = I_2$$

$$I_{D_2} = I_2 - I_1 = 3.32\text{ mA} - 0.212\text{ mA} \cong \mathbf{3.11\text{ mA}}$$

AND/OR Gates

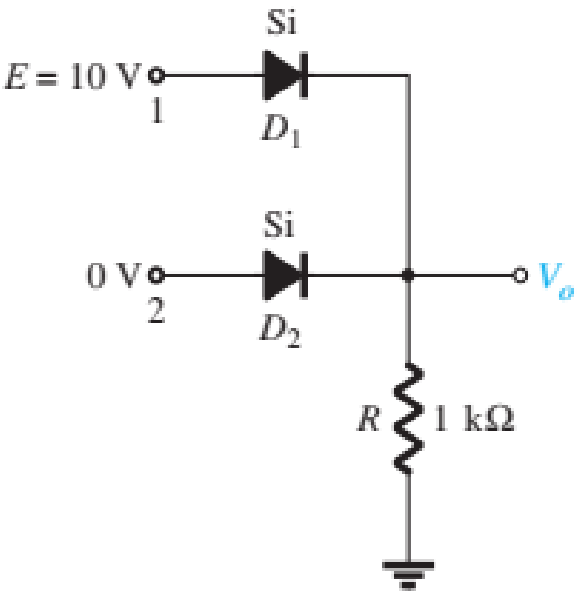


FIG. 2.39

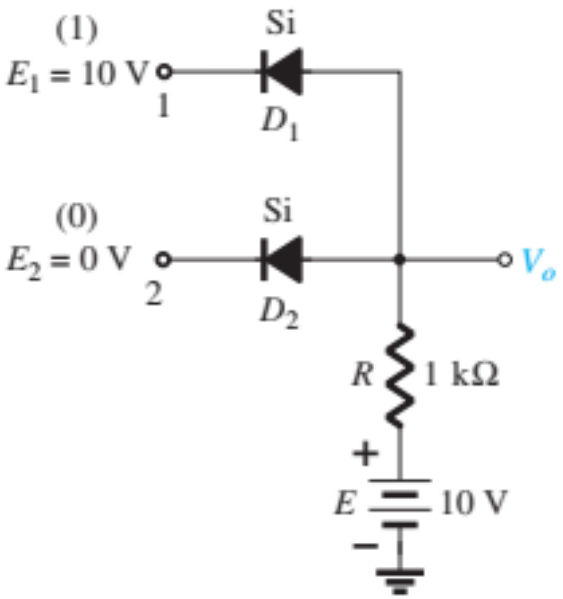


FIG. 2.42

AND/OR Gates

EXAMPLE 2.14 Determine V_o for the network of Fig. 2.39.

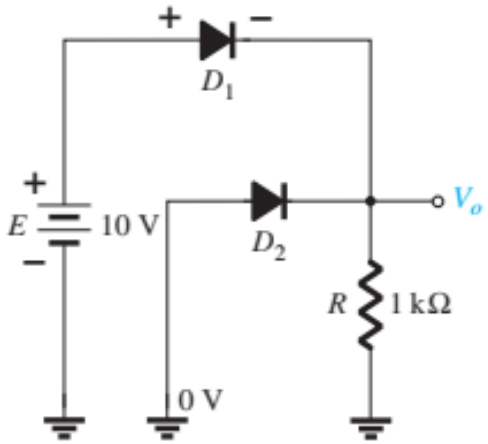


FIG. 2.40
Redrawn network of Fig. 2.39.

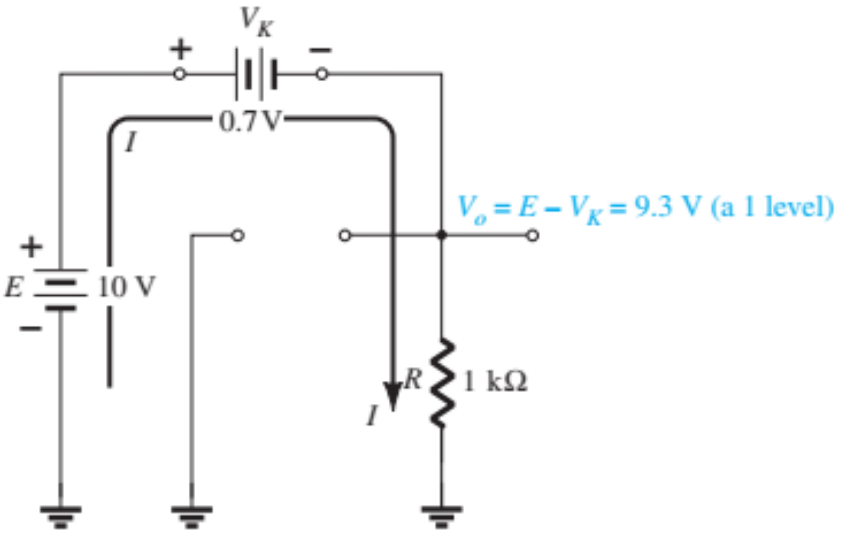


FIG. 2.41
Assumed diode states for Fig. 2.40.

$$I = \frac{E - V_D}{R} = \frac{10\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \mathbf{9.3\text{ mA}}$$

AND/OR Gates

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

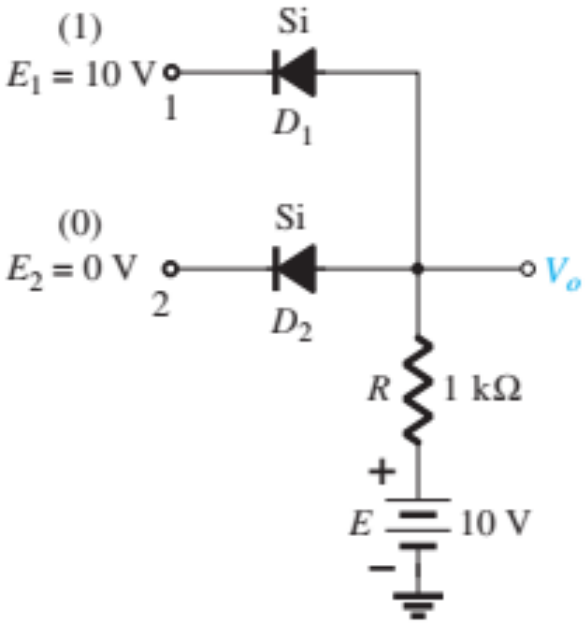


FIG. 2.42
Positive logic AND gate.

AND/OR Gates

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

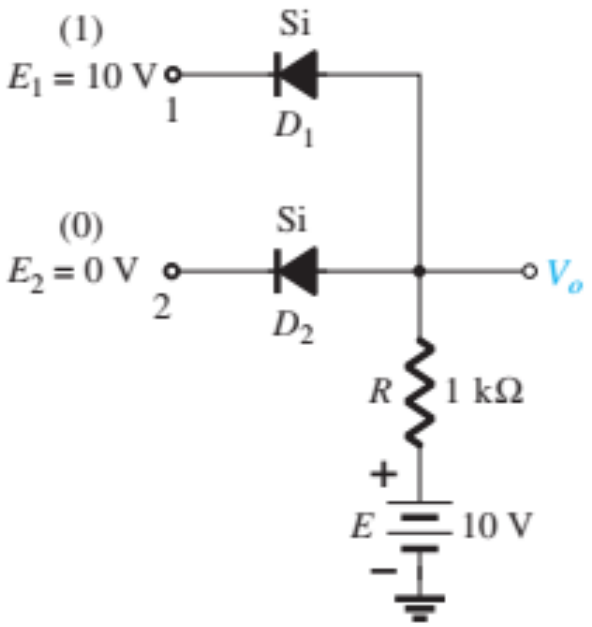


FIG. 2.42
Positive logic AND gate.

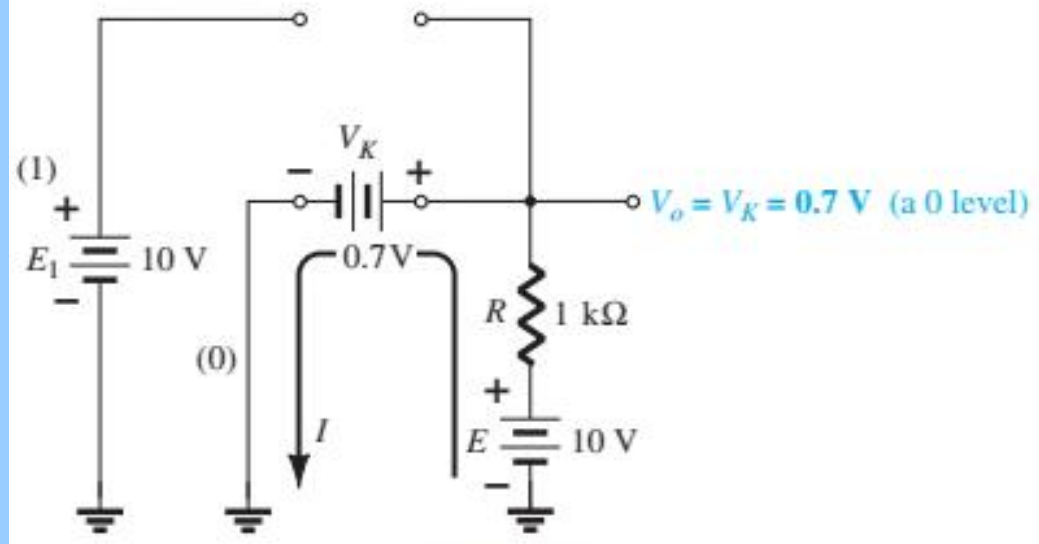


FIG. 2.43
Substituting the assumed states for the diodes of Fig. 2.42.

AND/OR Gates

EXAMPLE 2.15 Determine the output level for the positive logic AND gate of Fig. 2.42. An AND gate is one where a 1 output is only obtained when a 1 input appears at each and every input.

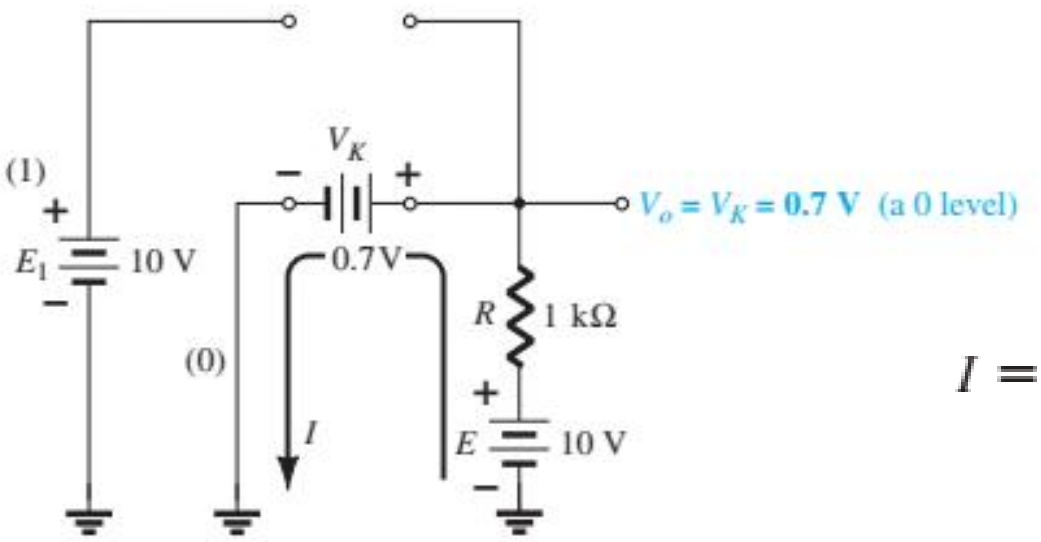


FIG. 2.43
Substituting the assumed states for the diodes of Fig. 2.42.

$$I = \frac{E - V_K}{R} = \frac{10\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \mathbf{9.3\text{ mA}}$$

Half-Wave Rectification

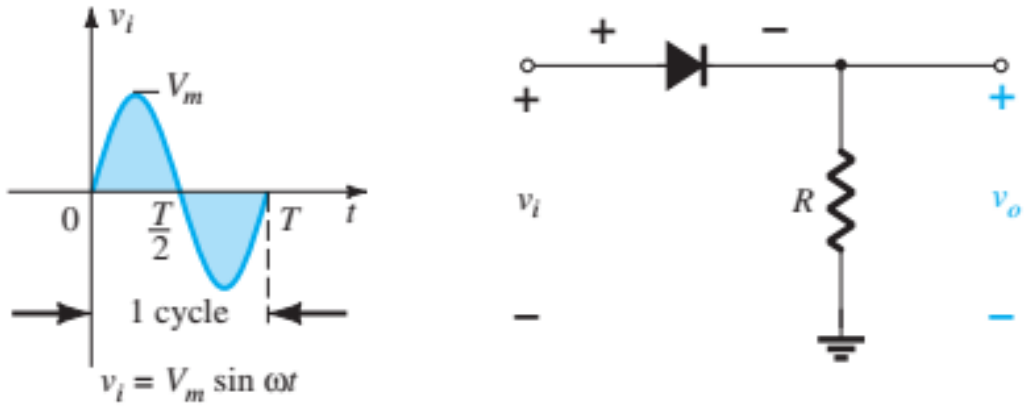


FIG. 2.44

Half-wave rectifier.

Half-Wave Rectification

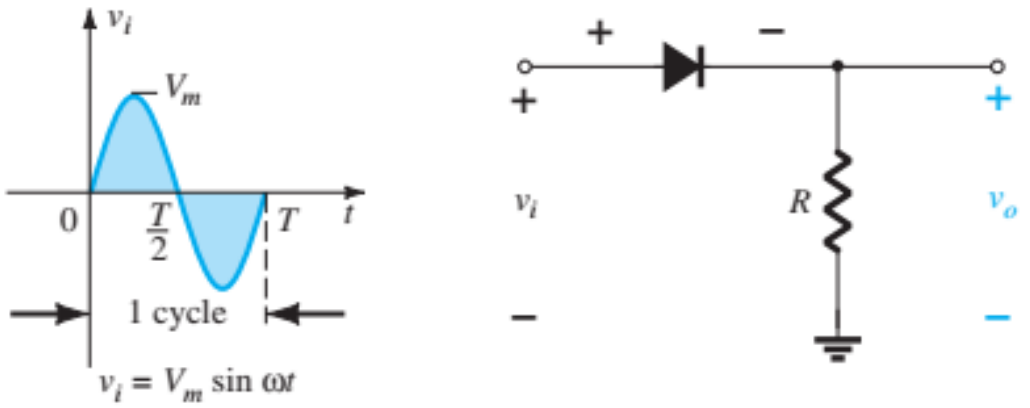


FIG. 2.44
Half-wave rectifier.

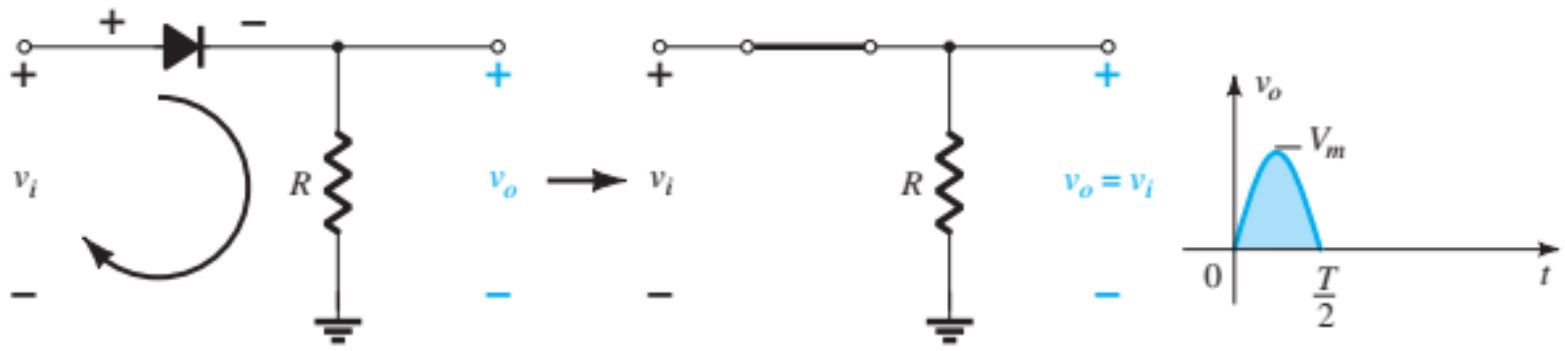


FIG. 2.45
Conduction region ($0 \rightarrow T/2$).

Half-Wave Rectification

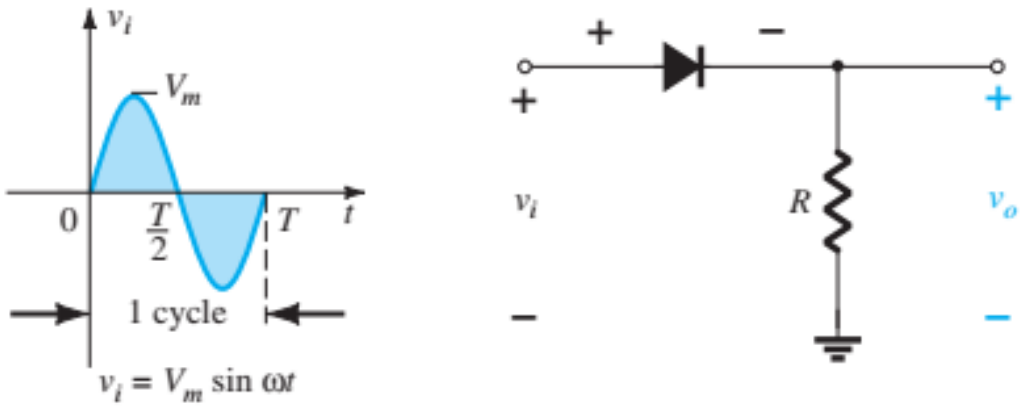


FIG. 2.44
Half-wave rectifier.

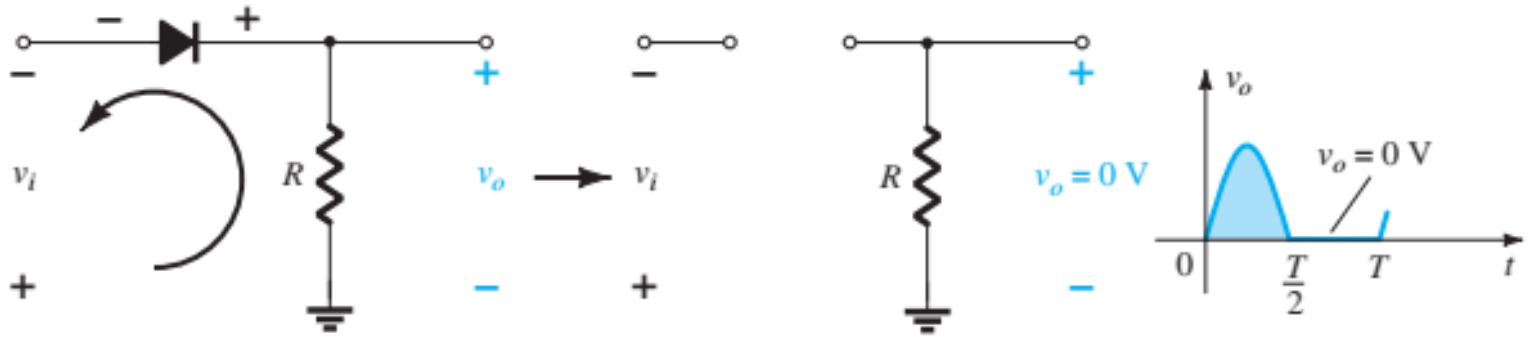


FIG. 2.46
Nonconduction region ($T/2 \rightarrow T$).

Half-Wave Rectification

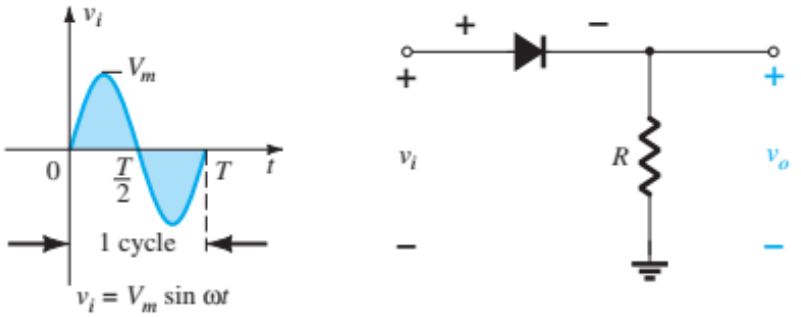


FIG. 2.44
Half-wave rectifier.

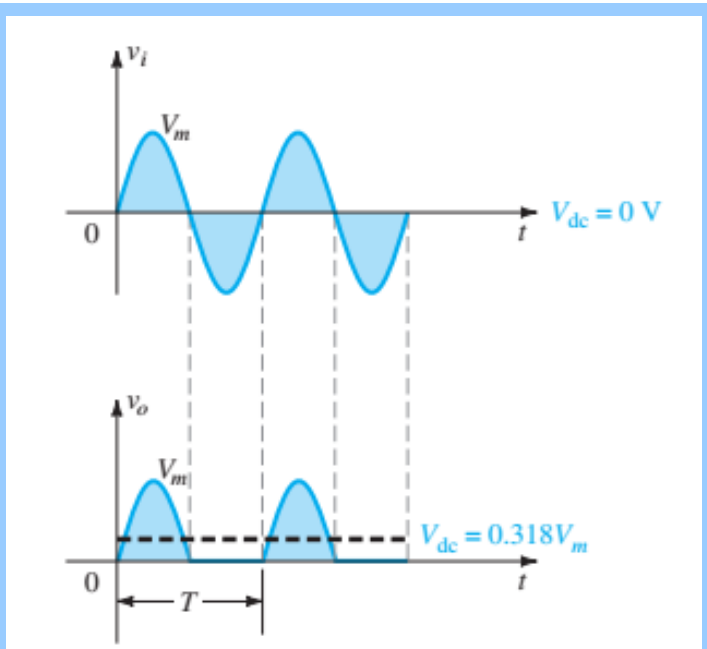


FIG. 2.47
Half-wave rectified signal.

Half-Wave Rectification

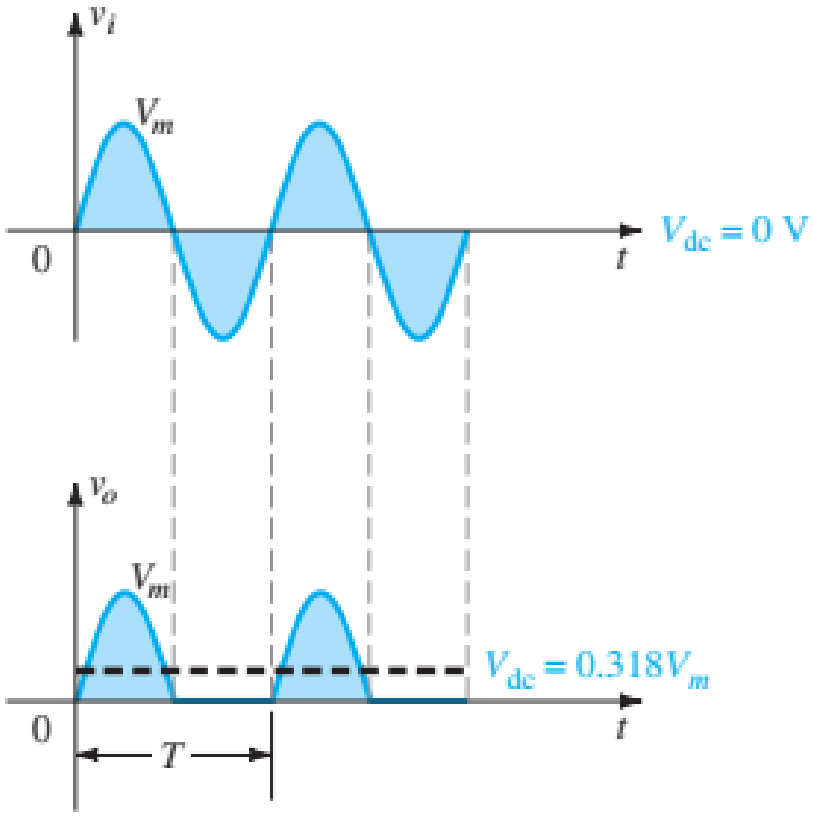


FIG. 2.47
Half-wave rectified signal.

$$V_{dc} = 0.318\,V_m$$ half-wave

Half-Wave Rectification

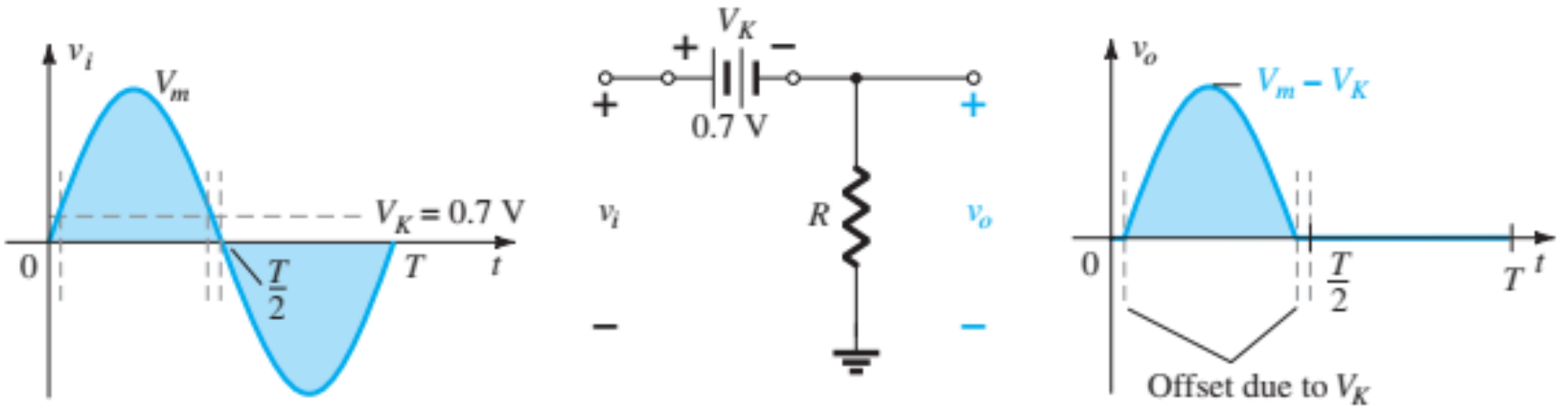


FIG. 2.48

Effect of V_K on half-wave rectified signal.

$$V_{dc} \cong 0.318(V_m - V_K)$$

where $V_m \gg V_K$,

Half-Wave Rectification

EXAMPLE 2.16

a. Sketch the output v_o and determine the dc level of the output for the network of Fig. 2.49.
b. Repeat part (a) if the ideal diode is replaced by a silicon diode.
c. Repeat parts (a) and (b) if V_m is increased to 200 V, and compare solutions using Eqs. (2.7) and (2.8).

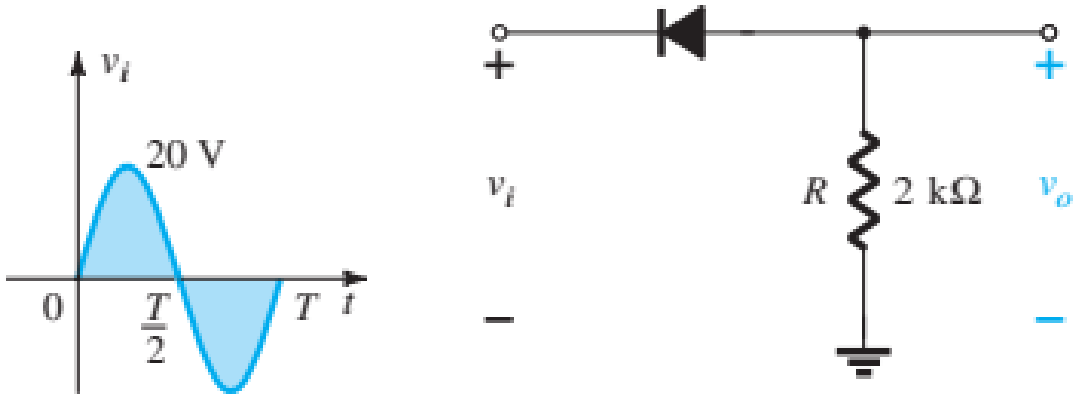
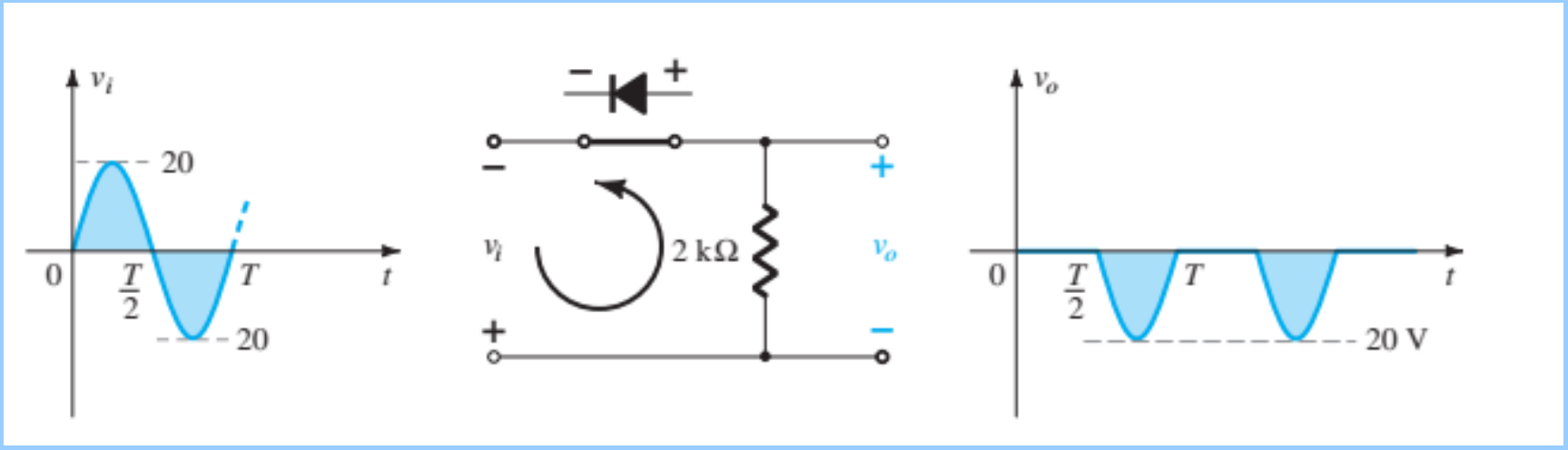


FIG. 2.49
Network for Example 2.16.

Half-Wave Rectification



a) $$V_{dc} = -0.318V_m = -0.318(20\text{ V}) = \mathbf{-6.36\text{ V}}$$

Half-Wave Rectification

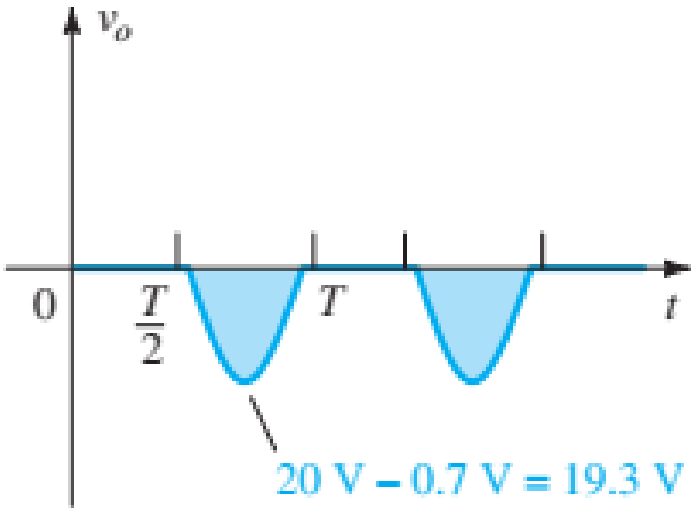


FIG. 2.51

Effect of V_K on output of Fig. 2.50.

b) $V_{dc} \cong -0.318(V_m - 0.7\text{ V}) = -0.318(19.3\text{ V}) \cong \mathbf{-6.14\text{ V}}$

Half-Wave Rectification

c for a) $V_{dc} = -0.318\,V_m = -0.318(200\text{ V}) = \mathbf{-63.6\text{ V}}$

c for b) $V_{dc} = -0.318(V_m - V_K) = -0.318(200\text{ V} - 0.7\text{ V})$

$= -(0.318)(199.3\text{ V}) = \mathbf{-63.38\text{ V}}$

Full-Wave Rectification

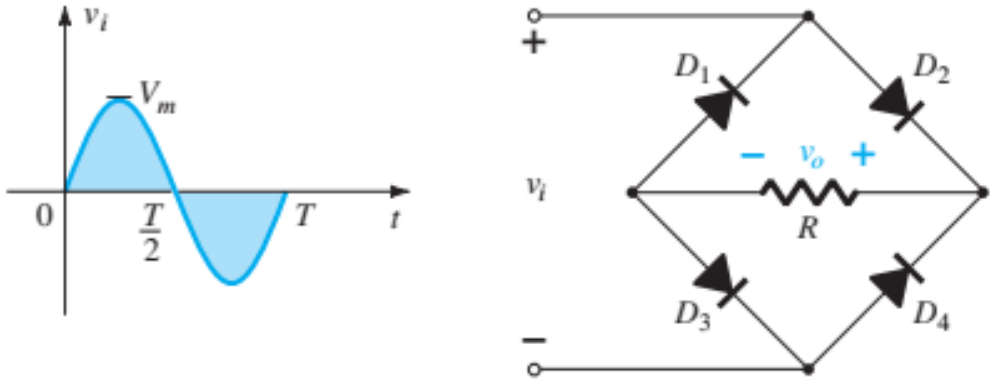


FIG. 2.53
Full-wave bridge rectifier.

Full-Wave Rectification

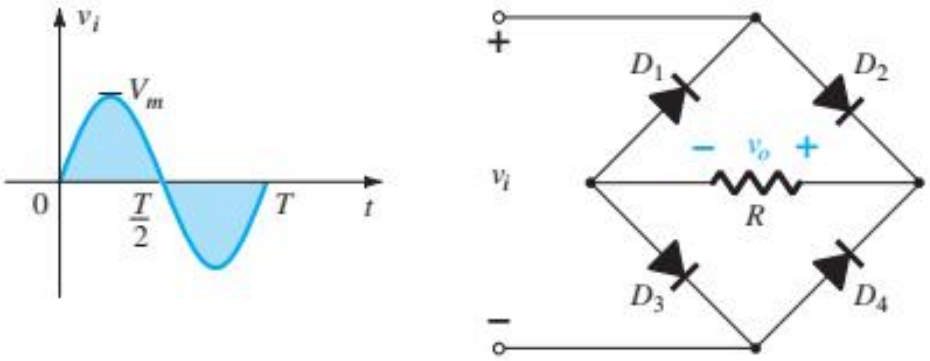


FIG. 2.53
Full-wave bridge rectifier.

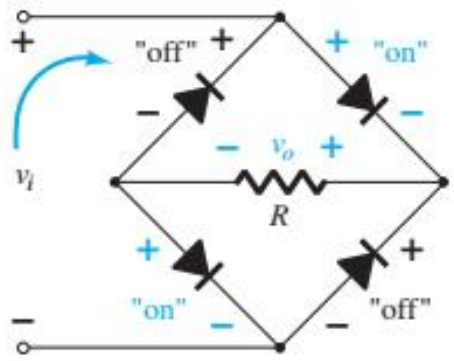


FIG. 2.54
Network of Fig. 2.53 for the period $0 \rightarrow T/2$ of the input voltage v_i.

Full-Wave Rectification

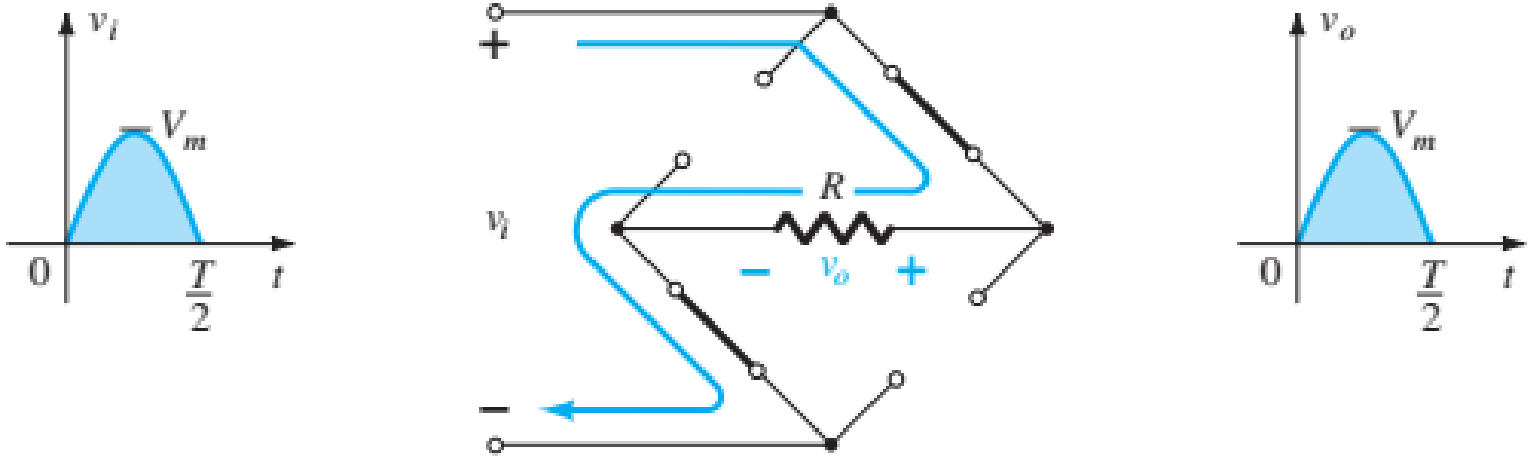


FIG. 2.55

Conduction path for the positive region of v_i.

Full-Wave Rectification

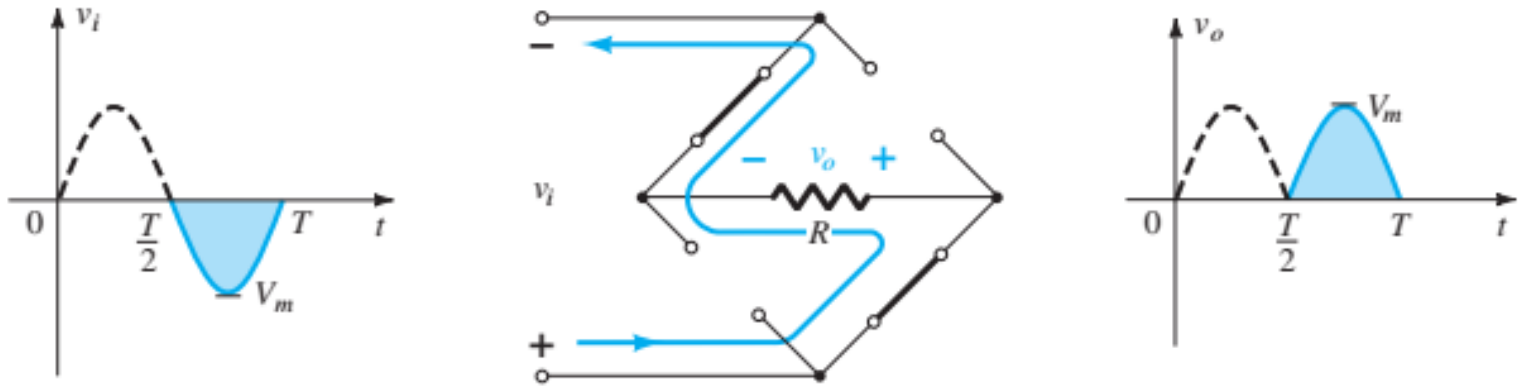
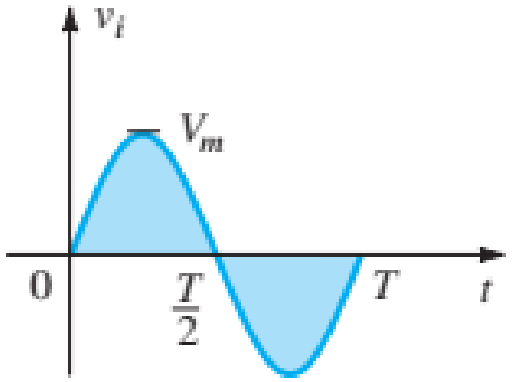


FIG. 2.56

Conduction path for the negative region of v_i.

Full-Wave Rectification



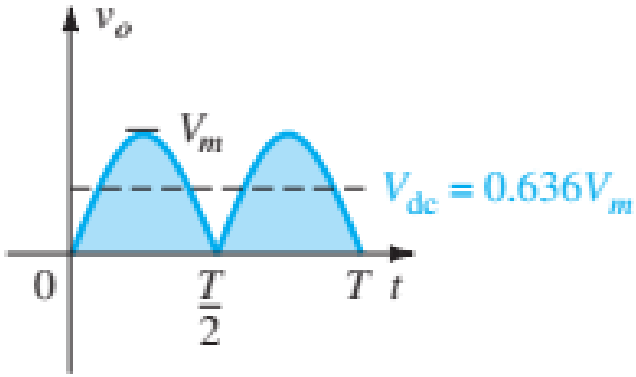


FIG. 2.57

Input and output waveforms for a full-wave rectifier.

Full-Wave Rectification

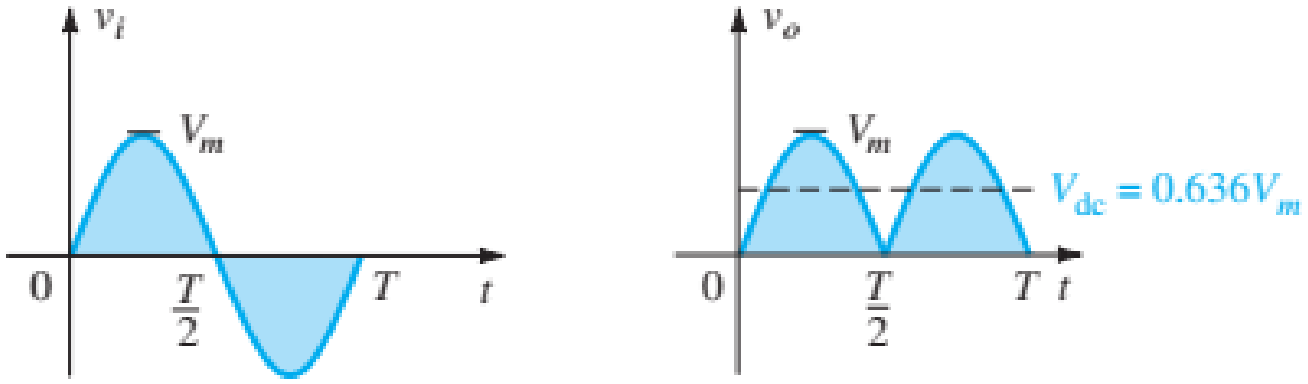


FIG. 2.57

Input and output waveforms for a full-wave rectifier.

$$V_{dc} = 2[\text{Eq. (2.7)}] = 2(0.318V_m)$$

$$\boxed{V_{dc} = 0.636\,V_m}$$ full-wave

Full-Wave Rectification

If silicon rather than ideal diodes are employed as shown in Fig. 2.58, the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$

and

$$v_o = v_i - 2V_K$$

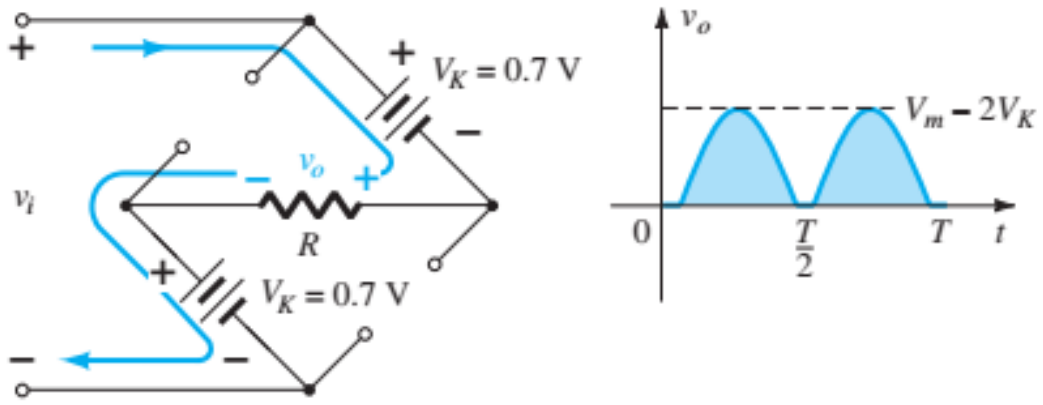


FIG. 2.58

Determining $V_{o_{max}}$ for silicon diodes in the bridge configuration.

$$V_{dc} \cong 0.636(V_m - 2V_K)$$

where $V_m \gg 2V_K$

Center-Tapped Transformer

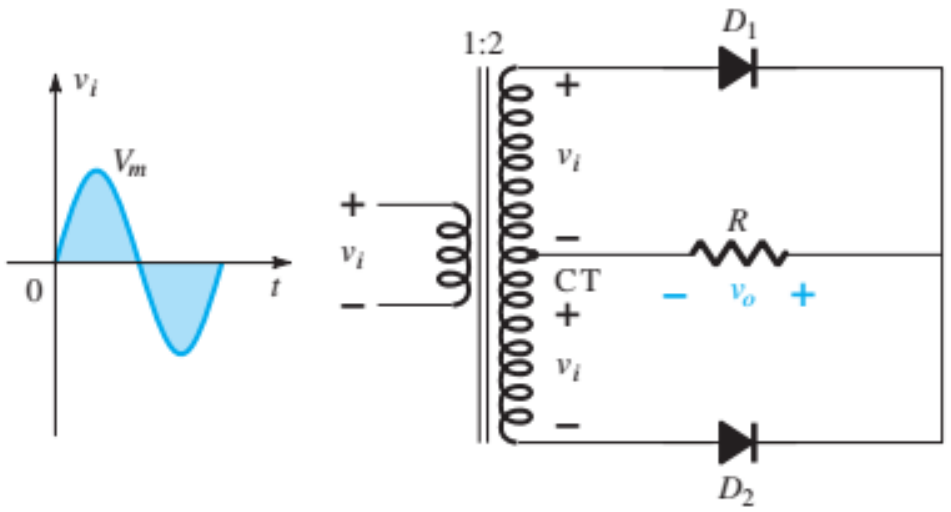


FIG. 2.60

Center-tapped transformer full-wave rectifier.

Center-Tapped Transformer

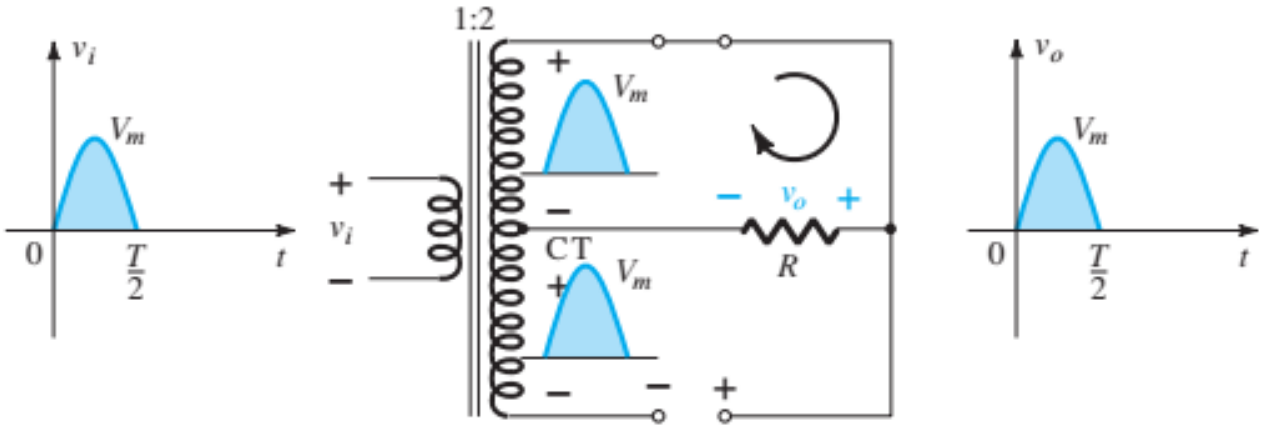


FIG. 2.61

Network conditions for the positive region of v_i.

Center-Tapped Transformer

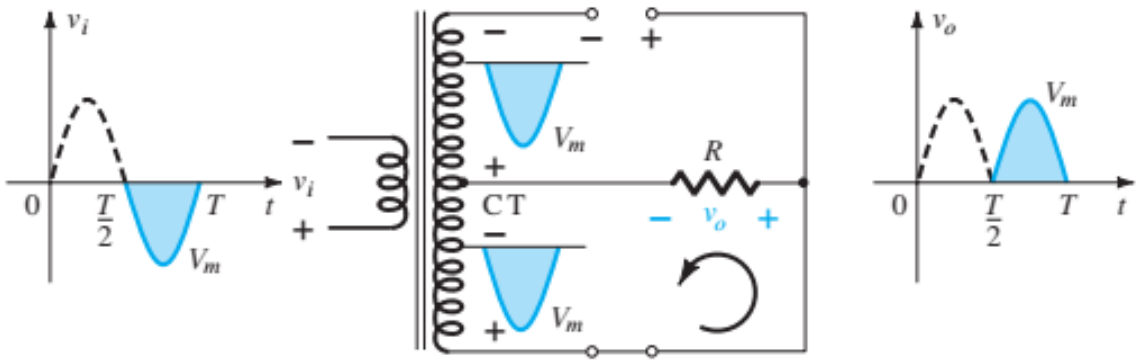


FIG. 2.62

Network conditions for the negative region of v_i.

Another Example

Determine the output waveform for the network of Fig. 2.64

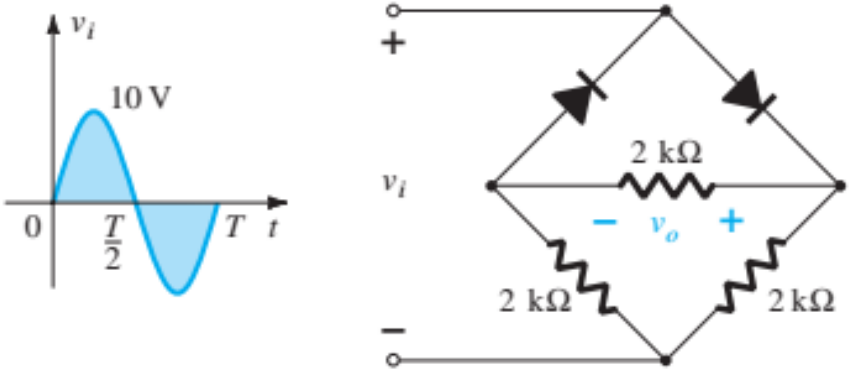


FIG. 2.64

Bridge network for Example 2.17.

Another Example

Determine the output waveform for the network of Fig. 2.64

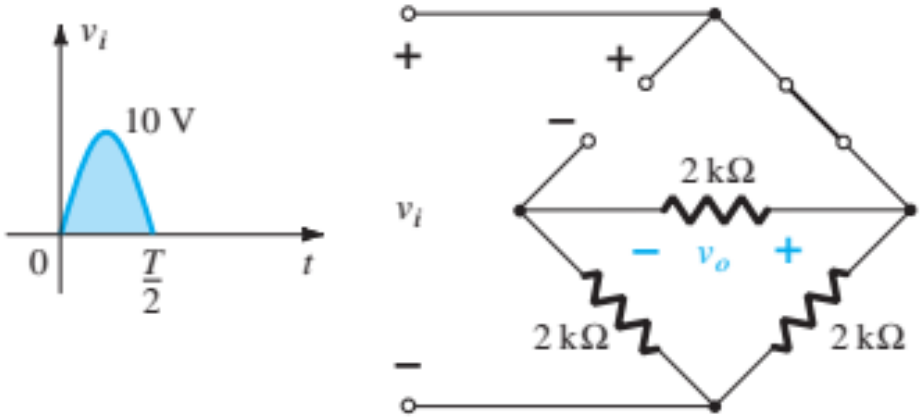


FIG. 2.65

Network of Fig. 2.64 for the positive region of v_i.

Another Example

Determine the output waveform for the network of Fig. 2.64

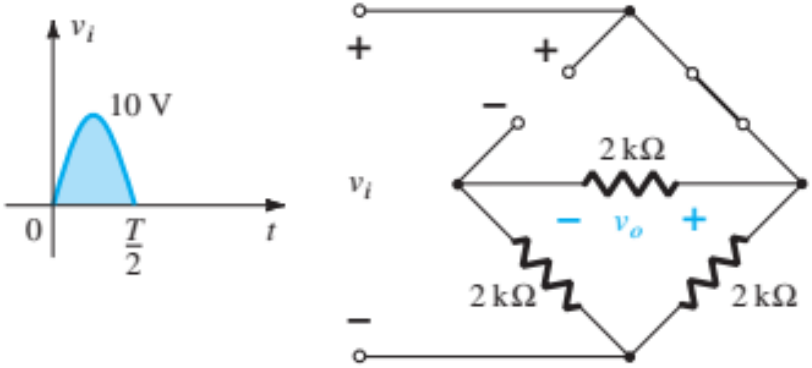


FIG. 2.65

Network of Fig. 2.64 for the positive region of v_i.

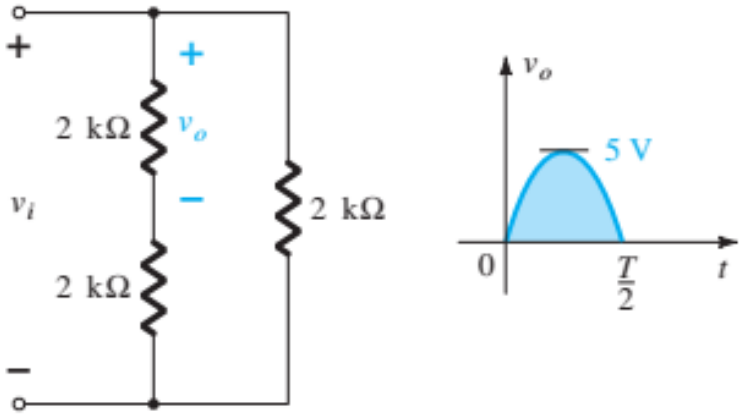


FIG. 2.66

Redrawn network of Fig. 2.65.

Another Example

Determine the output waveform for the network of Fig. 2.64

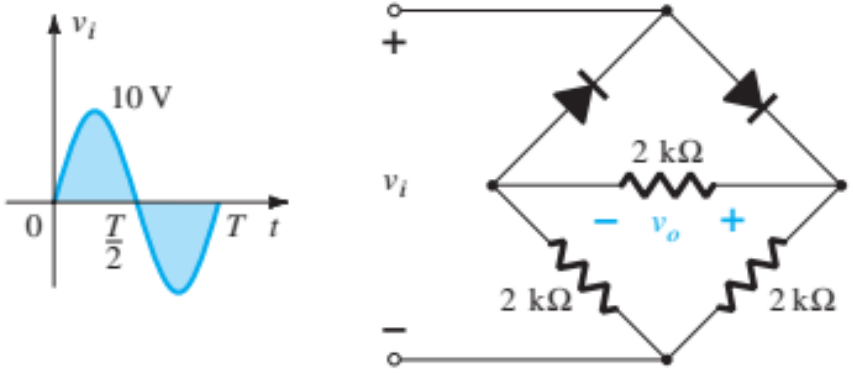


FIG. 2.64
Bridge network for Example 2.17.

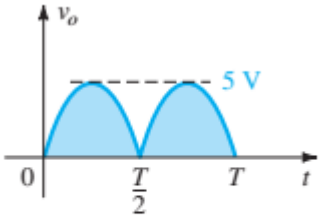


FIG. 2.67
Resulting output for Example 2.17.

$$V_{dc} = 0.636(5\text{ V}) = 3.18\text{ V}$$